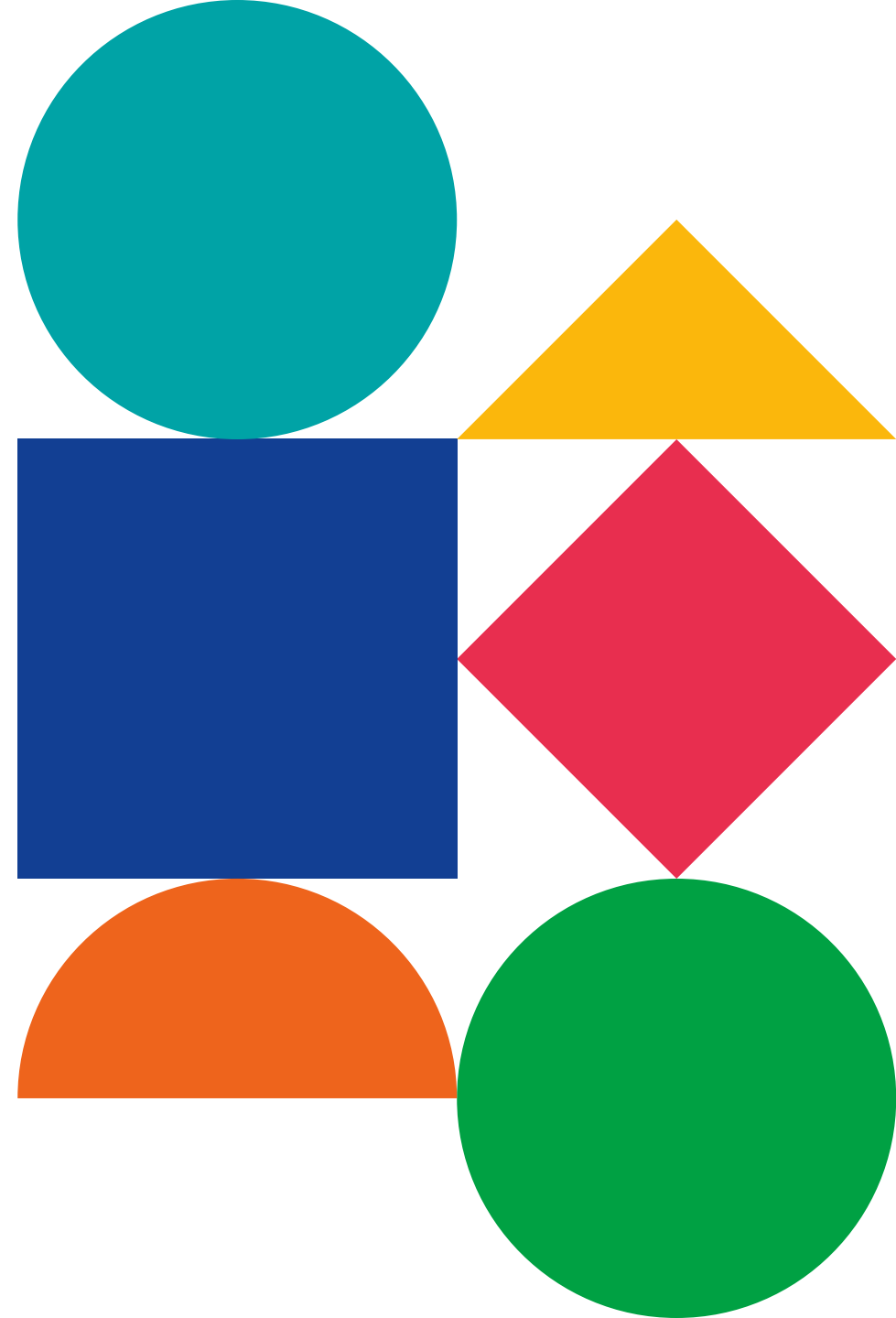


Versioning with Git & GitHub

For code and documentation - Day 2

Bruna Piereck

Dec 2025



REVIEW QUIZ

 **Kahoot!**



Experimenting Risk Free

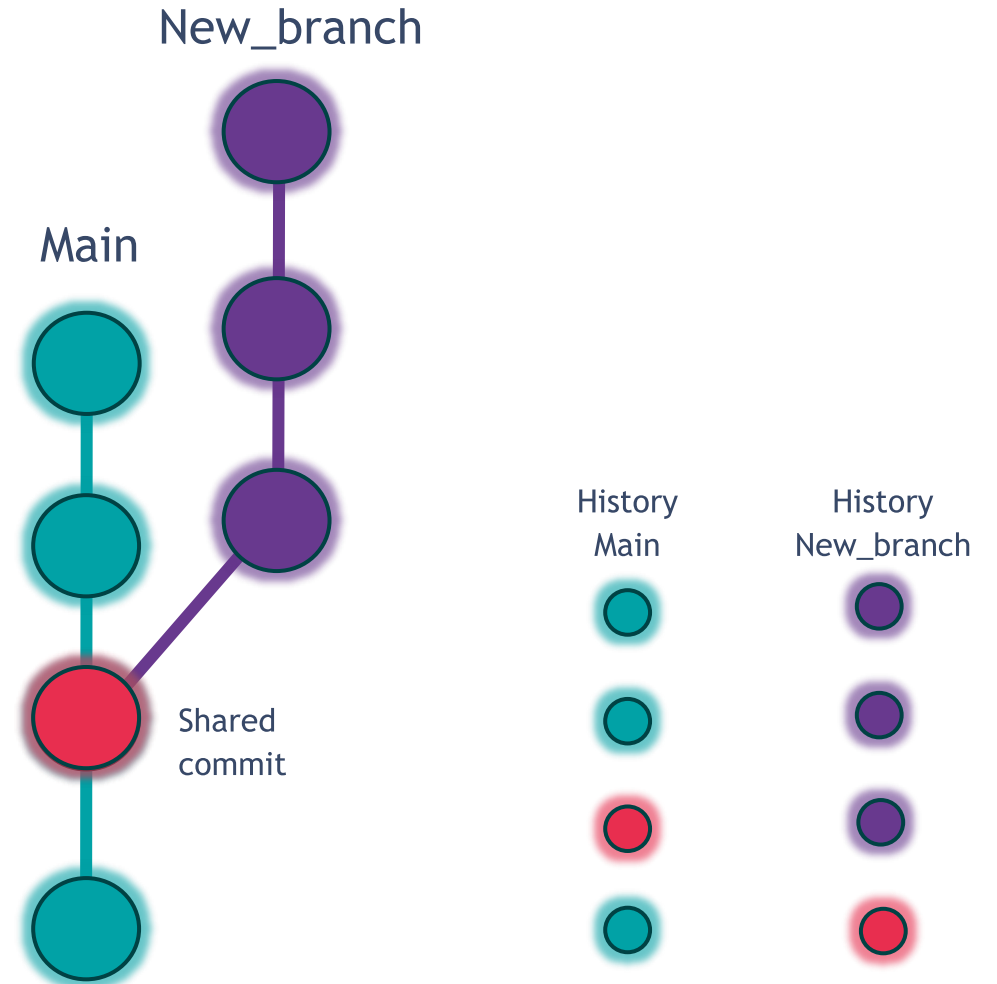
Alternative
timeline

Alternative history

```
user$ git branch <name>
```

Branch

- Has a specific name
- Alternative history of commits
 - Independent timeline
- Shared commit between timelines



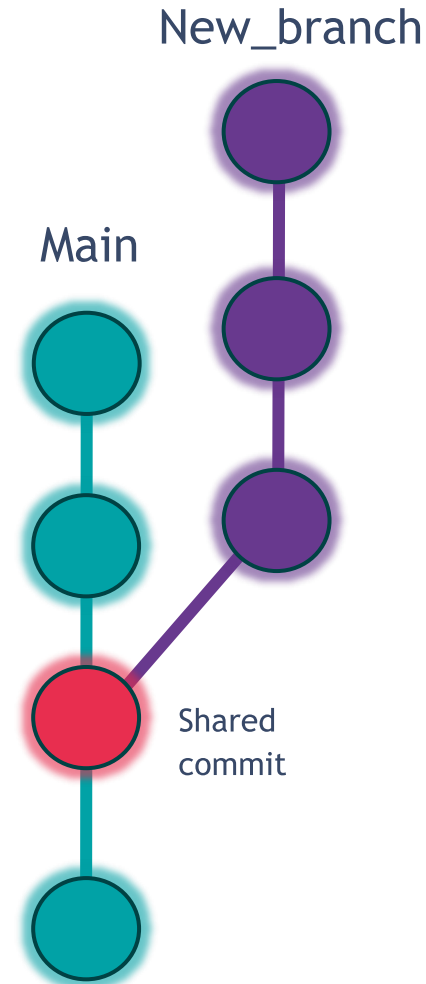
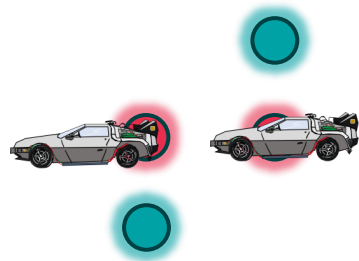
Alternative history

```
user$ git branch <name>
```

Utility

1. Experiment risk free
2. Collaborating in a project
3. Work in different tasks
4. Testing different possible solutions

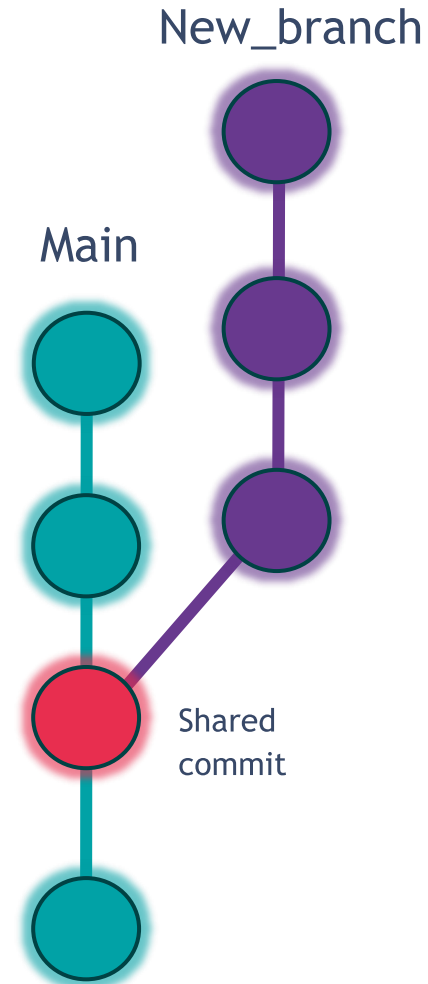
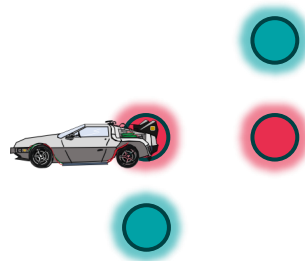
P.s.: Creating $\neq$ Working on



Moving between timelines

```
user$ git checkout <branch_name>
```

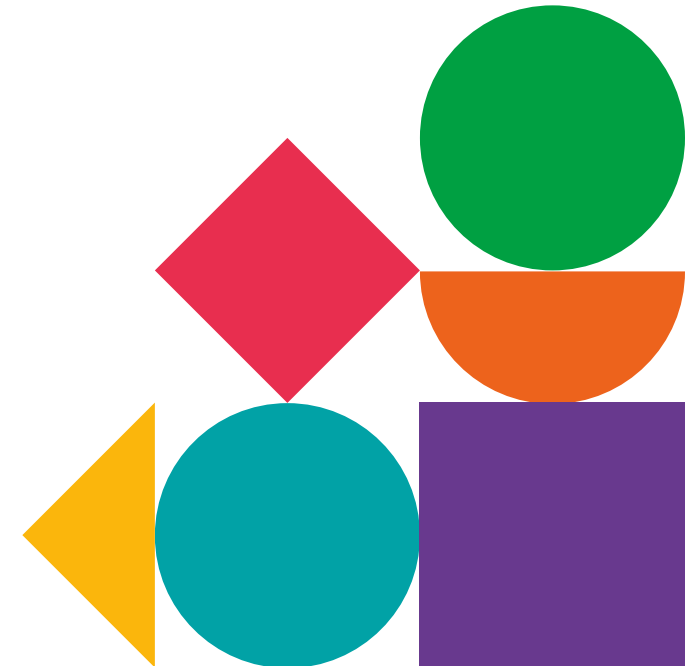
- Move between branches
- Move between commits
(!! Dangerous)



ACTIVITY

4. Collaborating & making use of branches

- Make sure you have a “working tree clean”
- Close all the files
- Go to your partner’s project
 - Sync all the content
 - Create your branch
`git branch <name>`
 - Checkout to your new branch
`git checkout <name>`
 - Write the concepts of Branching & Checkout
 - Write why branching can be useful
 - Commit and Push



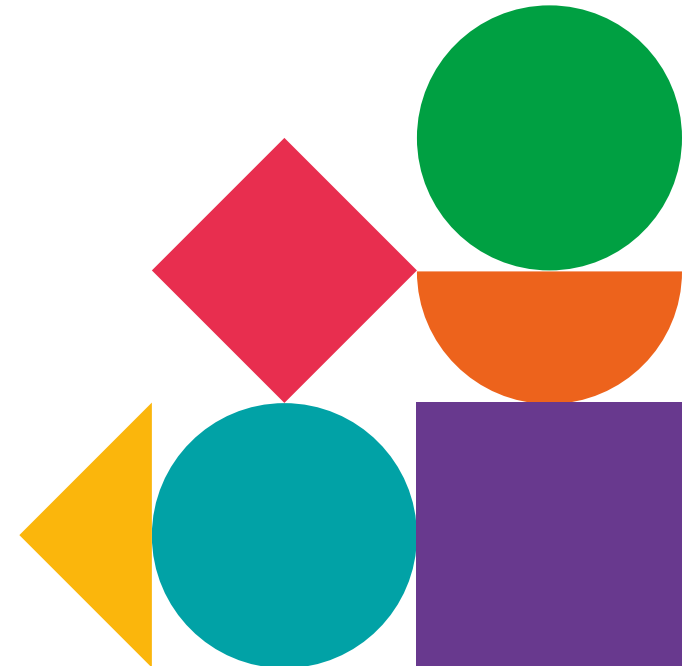
New branches & collaborations

```
user$ git branch -- list
```

- Check local branches

```
user$ git branch -a
```

- Check all branches
 - Local
 - Remote



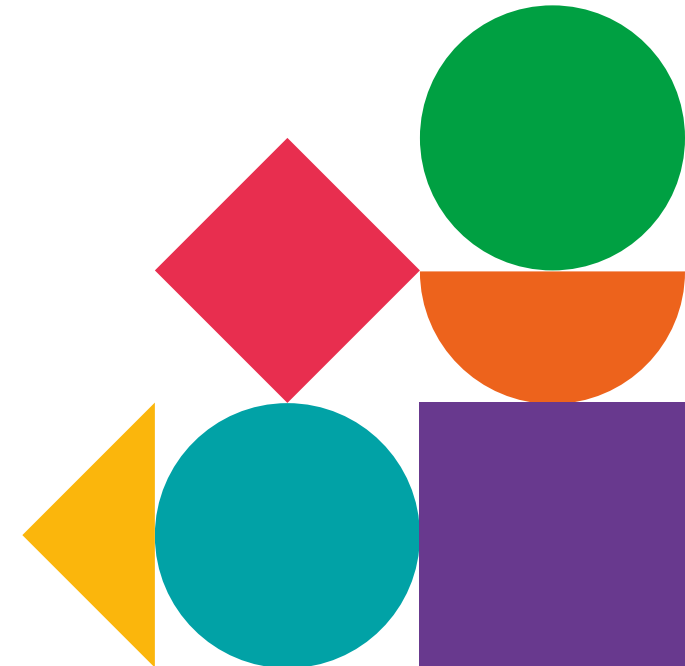
ACTIVITY

2. Go back to your project

- Sync the content
- Check the updates in the new branch your collaborator has created

```
user$ git branch - a
```

```
user$ git checkout <name>
```

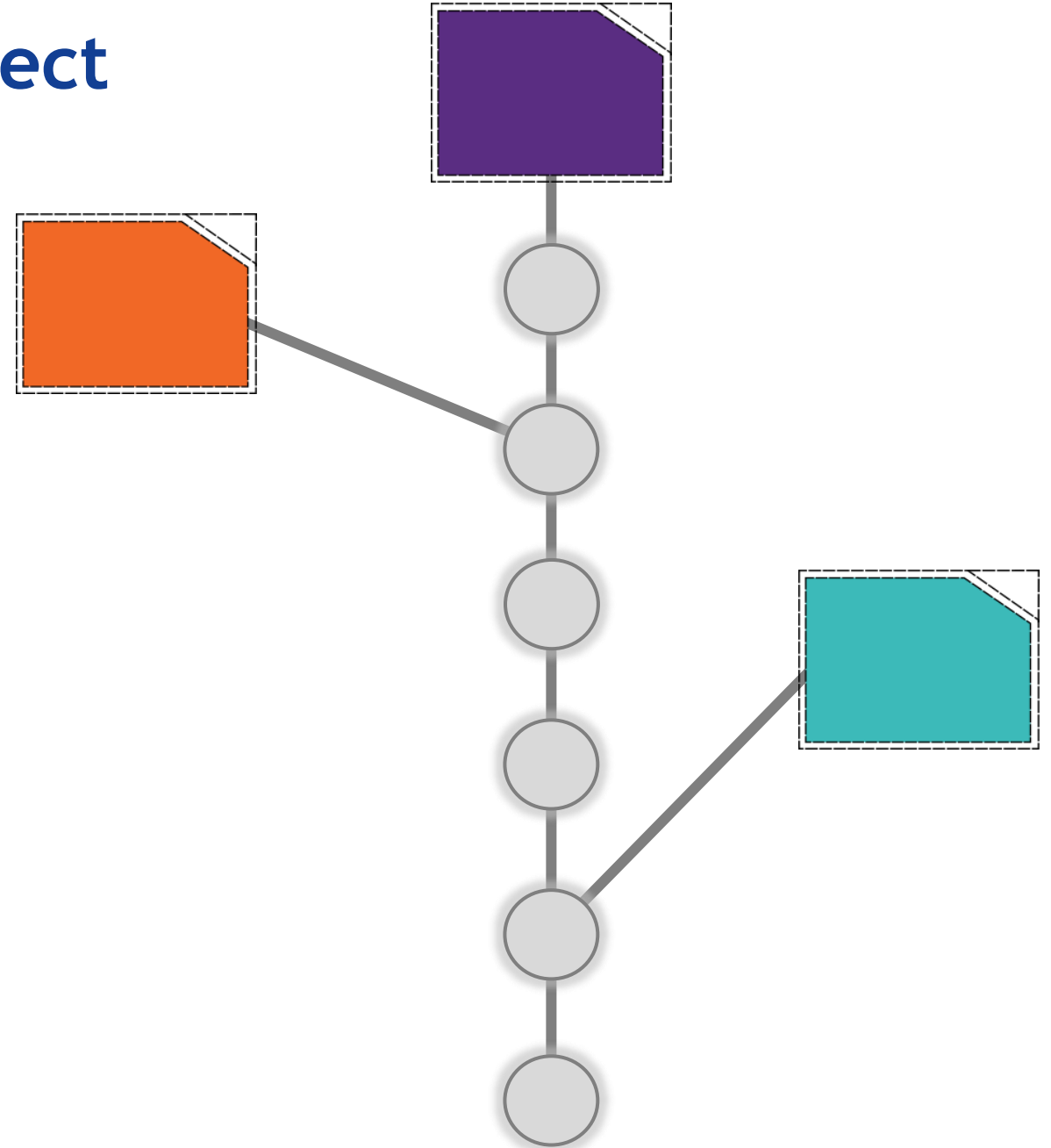
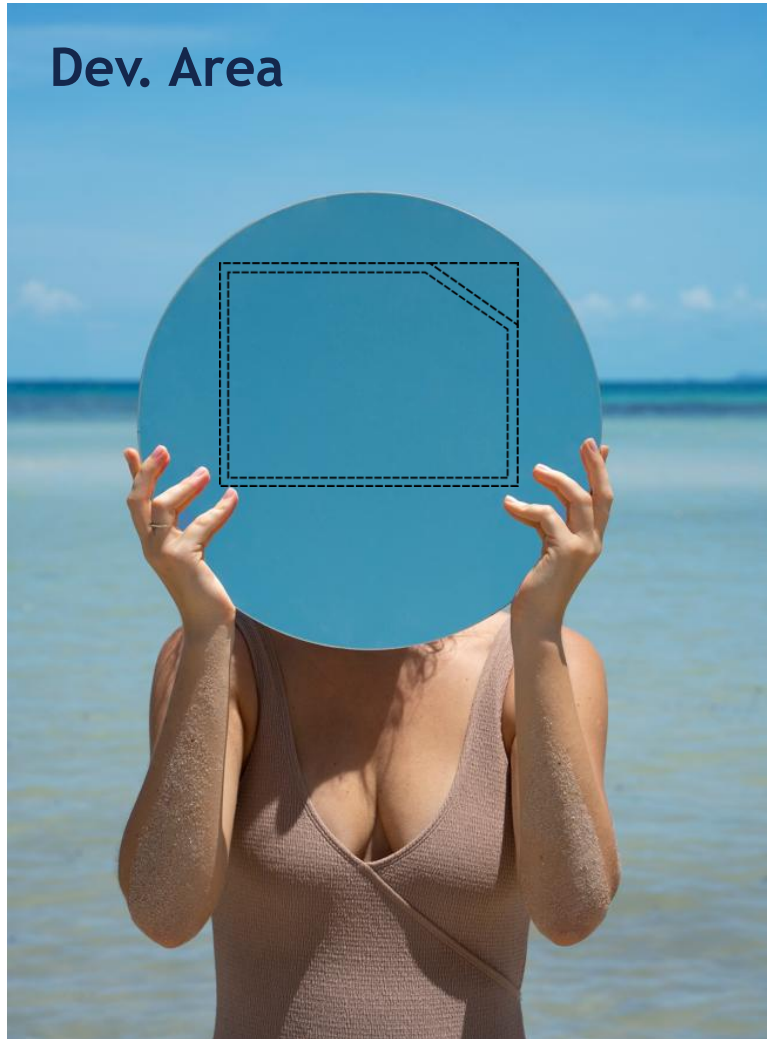


The “Mirror” effect

Dynamic
Dev. area

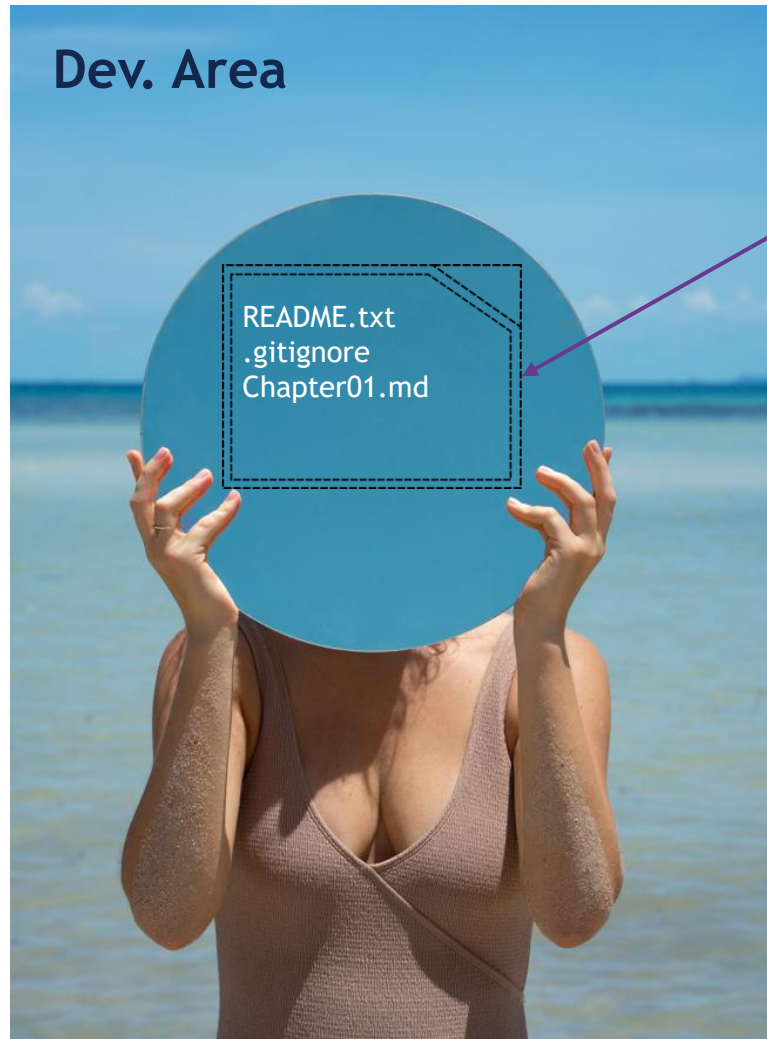


Branches: The “mirror” effect

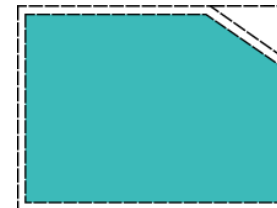
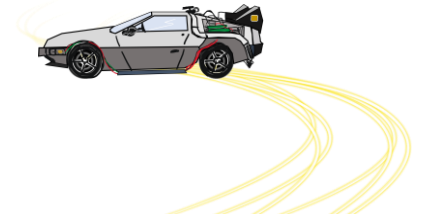


Branches: The “mirror” effect

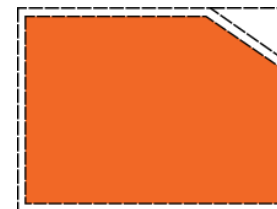
Git checkout vib_green



README.txt
.gitignore
Chapter01.md



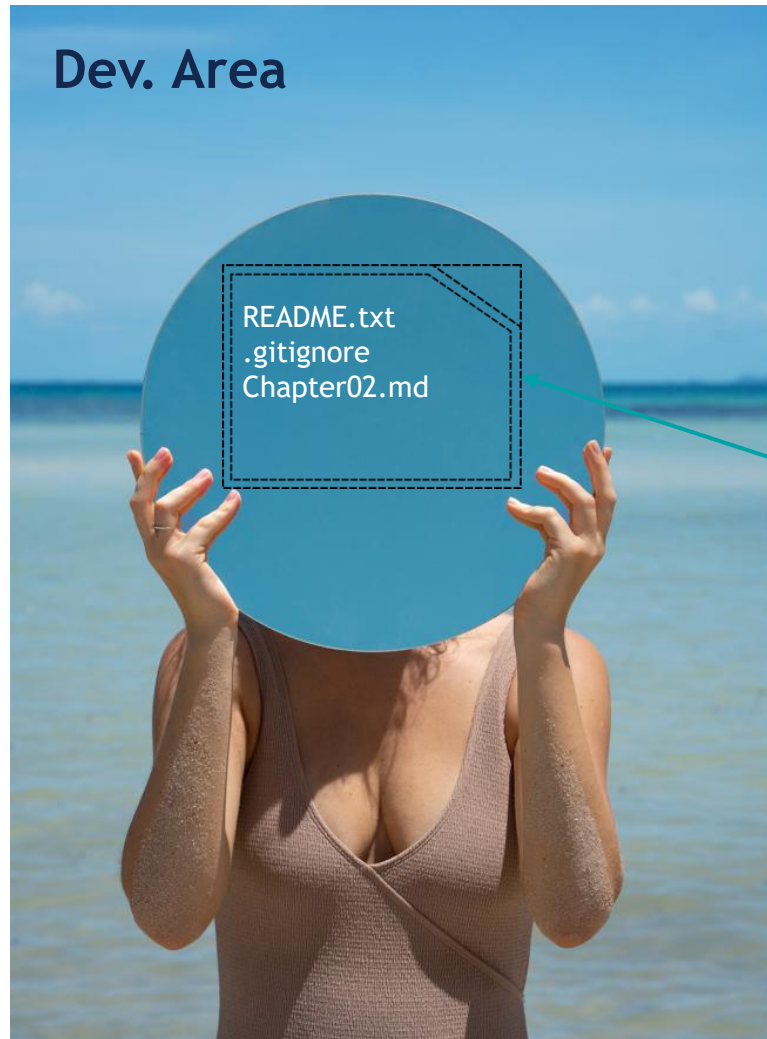
README.txt
.gitignore
Chapter02.md



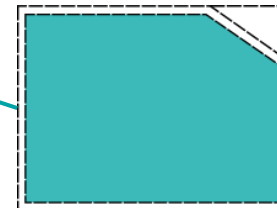
README.txt
.gitignore
Chapter01.md
Illustrations

Branches: The “mirror” effect

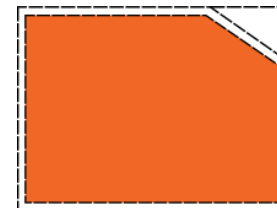
Git checkout vib_green



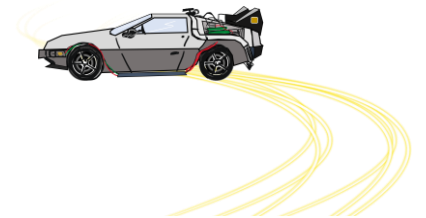
README.txt
.gitignore
Chapter01.md



README.txt
.gitignore
Chapter02.md

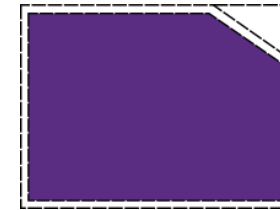


README.txt
.gitignore
Chapter01.md
Illustrations

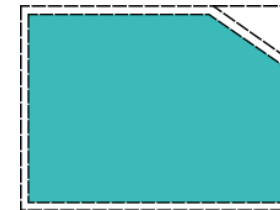


Branches: The “mirror” effect

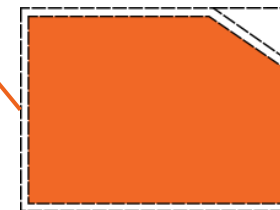
Git checkout vib_green



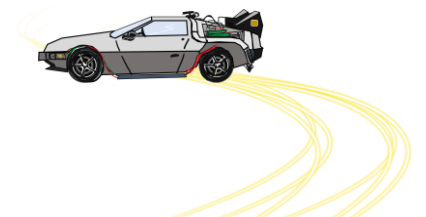
README.txt
.gitignore
Chapter01.md



README.txt
.gitignore
Chapter02.md



README.txt
.gitignore
Chapter01.md
Illustrations



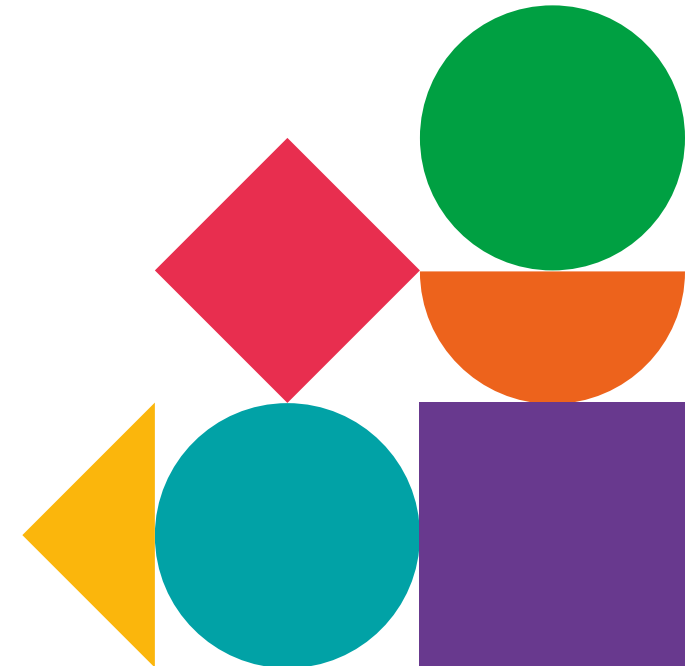


Coffee time

ACTIVITY

3. Compare content in both timelines

- Check your branch history
- Close all files
- Checkout to the main branch
`git checkout main`
- Check the history
 - Is there any differences?
- Open the files
 - Is there any information missing?
 - Why?



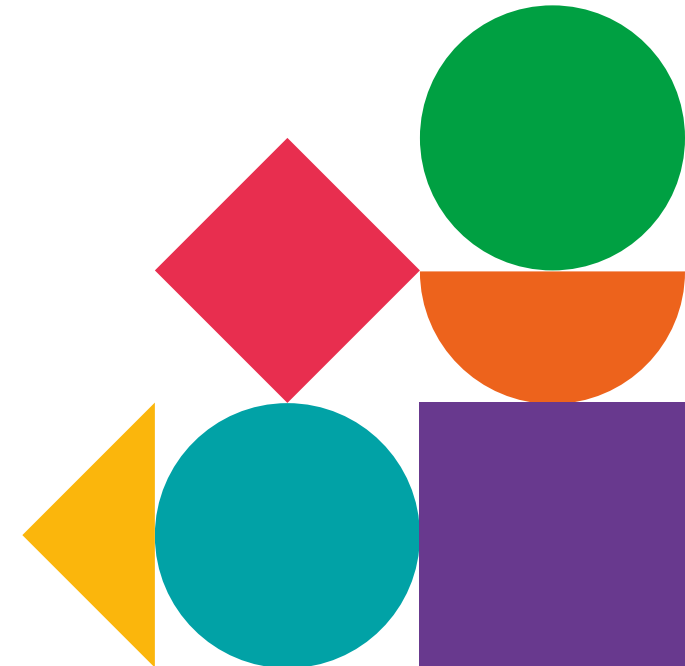
ACTIVITY

4. Working with branches

- Create a new_branch
- Write about the “mirror” effect
- Commit & Push

5. Comparing

- Close all files
- Move back to main
 - Is the content there?
 - Why?
- Can you see all branches in GitHub?



QUIZ

■ Why I can't see the changes in the file present in the Main Branch?

A. I lost what I just did?

B. It can only be found in the branch where I edited it

C. Git reloads my files based in the branch I am in

 B and C are correct

E. I have no idea



QUIZ

- How many copies of each file I have in the Dev. Area?

 Only 1.

B. I have 2, one for each branch

C. As many as branches I have

D. As many as versions I have

- Where are the other versions?

A. In another folder

B. I GitHub only

 In the .git folder



QUIZ

■ Where can I create a branch?

A. Only projects I own

☒ B. Projects I own or collaborate

C. Any repository

D. B and C are correct



QUIZ

- Can you see all branches your collaborators have created?

-  Only the ones pushed to GitHub
- B. Yes, I can see all
- C. No, I can't see them unless invited for collaboration
- D. I can see them on GitHub always





Change the past at your own risk

Checkout to an
old commit



Detached head

```
user$ git checkout <commit_ID>
```

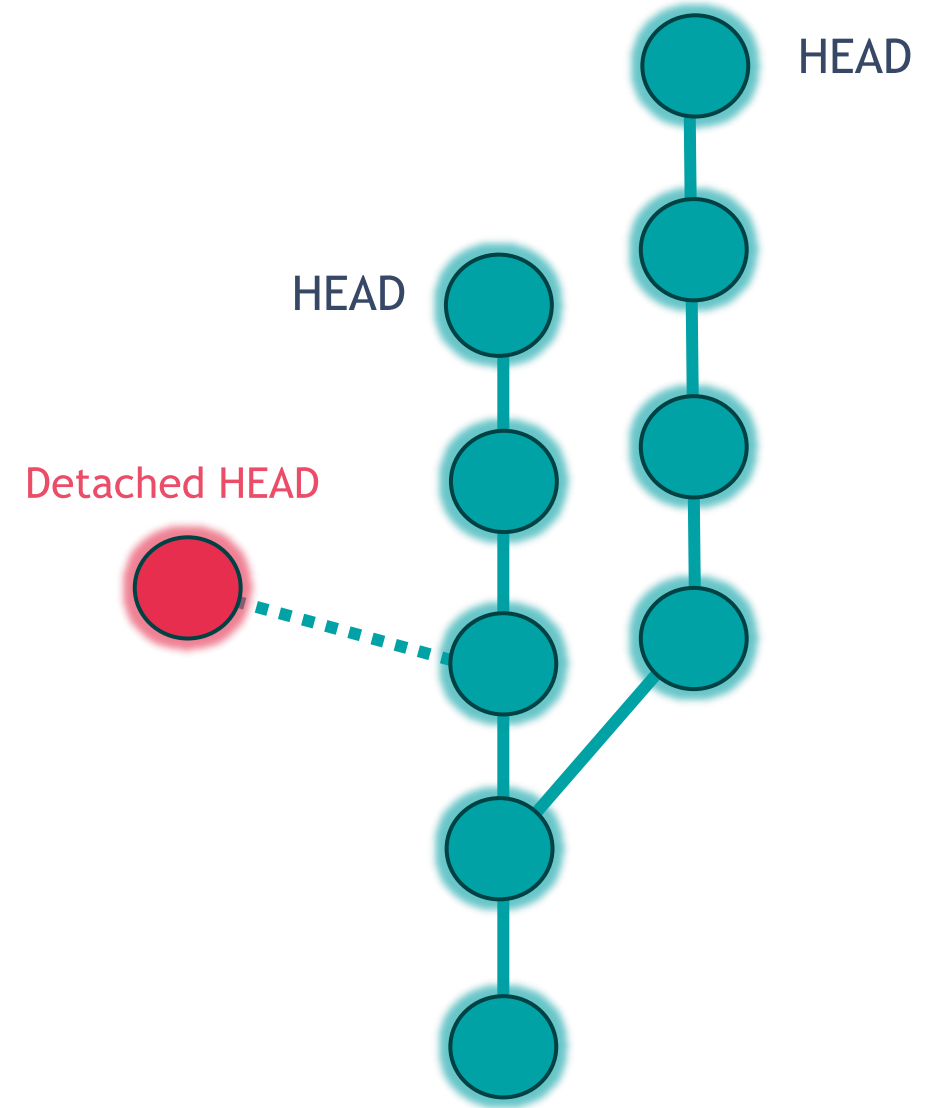
- You can't change the past
- You CAN create “an alternative future”
 - Be careful !!

You are in 'detached HEAD' state. You can look around, make experimental changes and commit them, and you can discard any commits you make in this state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may do so (now or later) by using `-c` with the switch command. Example:

```
git switch -c <new-branch-name>
```

```
user$ git switch -c <name_new_branch>
```



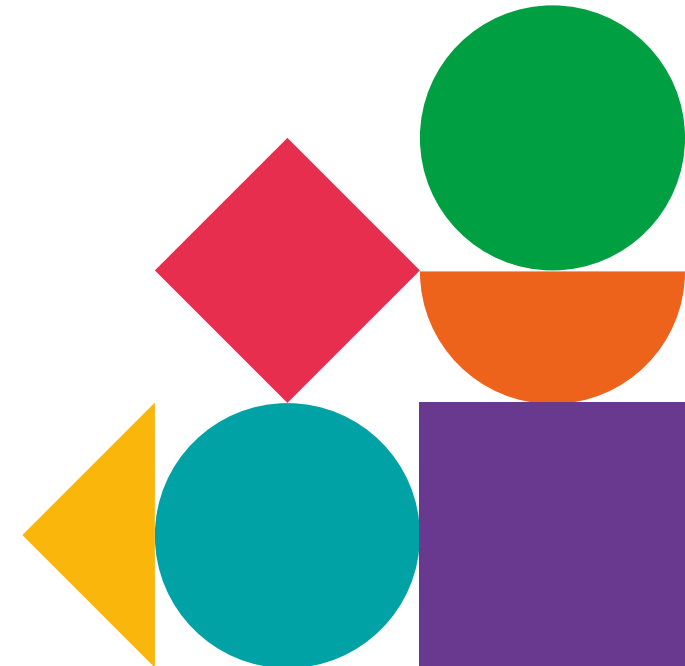
ACTIVITY

6. Changing the past

- Move to an old commit
- Add some text (2 steps)
 - Commit without switching
 - *Check in the list of branches*
 - Commit and do switch - c
 - *Check the list of branches*
- What happens?

```
user$ git checkout <commit_ID>
```

```
user$ git switch -c <name_new_branch>
```



Branches review

```
user$ git branch -- list
```

```
-a
```

```
<name>
```

List the local branches

List all branches, local & remote

Create a new branch

```
user$ git checkout <name / ID >
```

```
-b <name>
```

Go to branch

Create and go to new branch

```
user$ git switch -c <name_new_branch>
```

Transform a detached HEAD in a branch

Merging

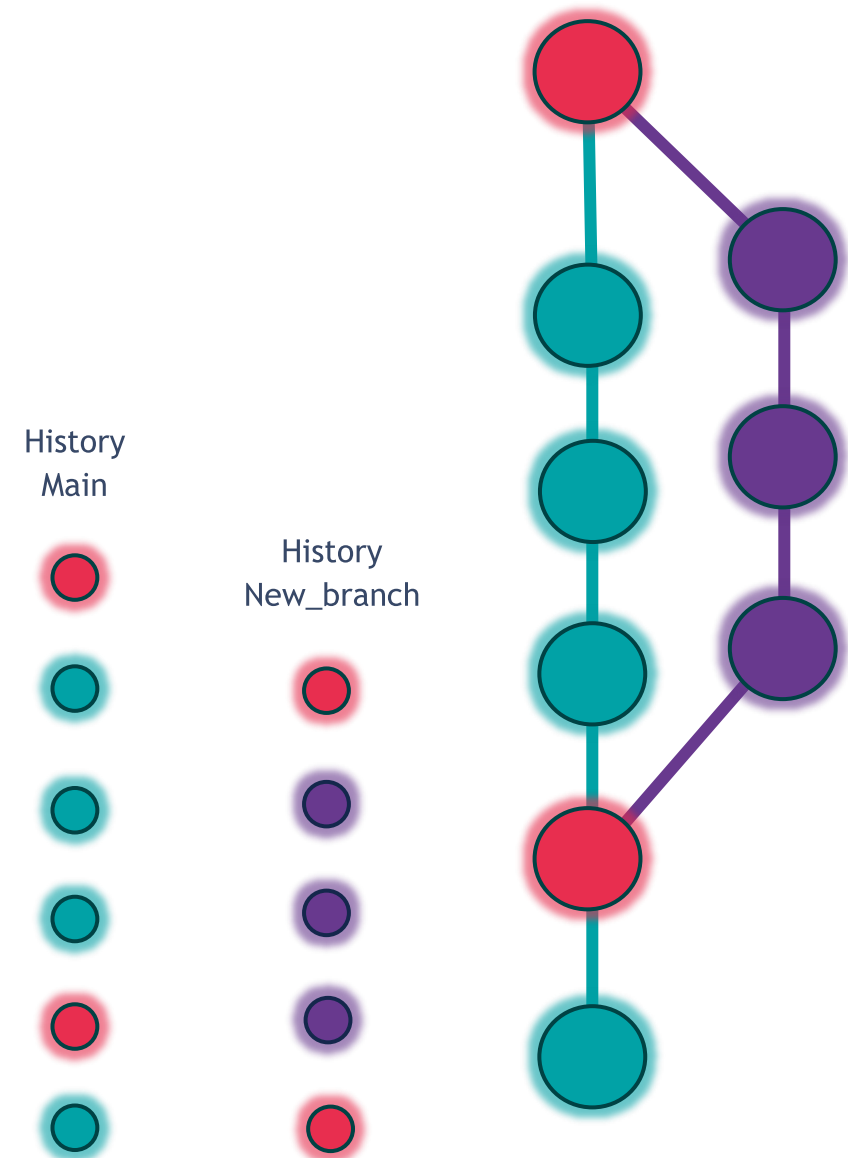
Branches come
together

Merging Branches

```
user$ git merge <branch_name>
```

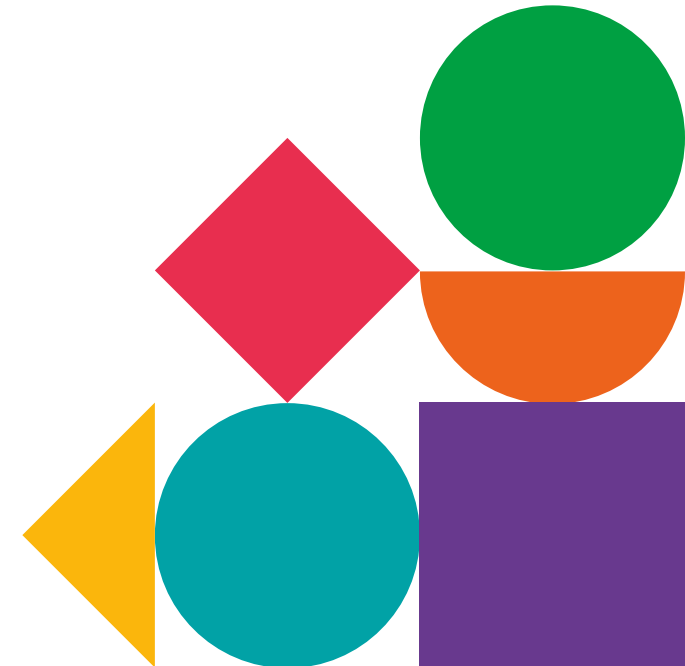
- Use cases
 1. Experiment worked out
 2. Finished collaboration
 3. Problem solved
- Too many branches
 - Clean up

```
user$ git branch -d <branch_name>
```



ACTIVITY

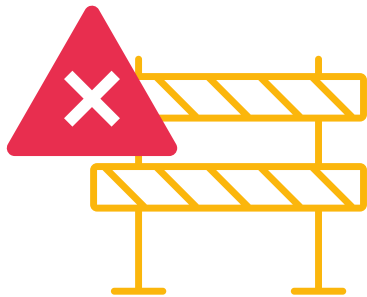
7. Create a new branch
 - Write about merging



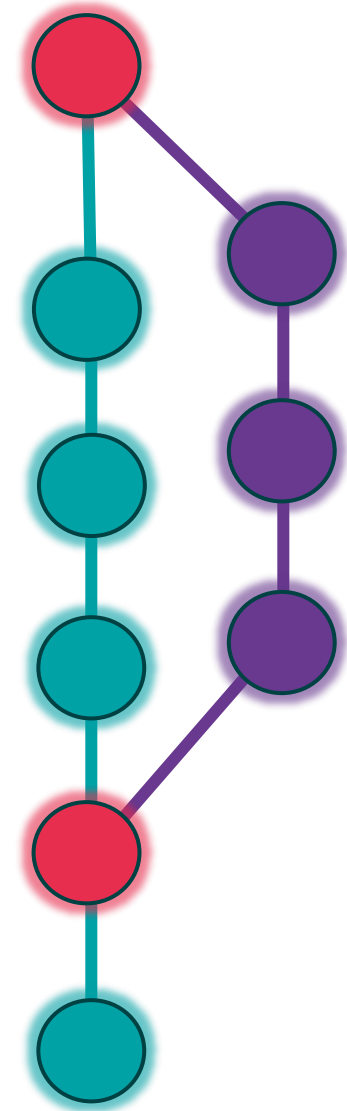
Merging Branches

```
user$ git merge <branch_name>
```

- Use cases
 1. Experiment worked out
 2. Finished collaboration
 3. Problem solved



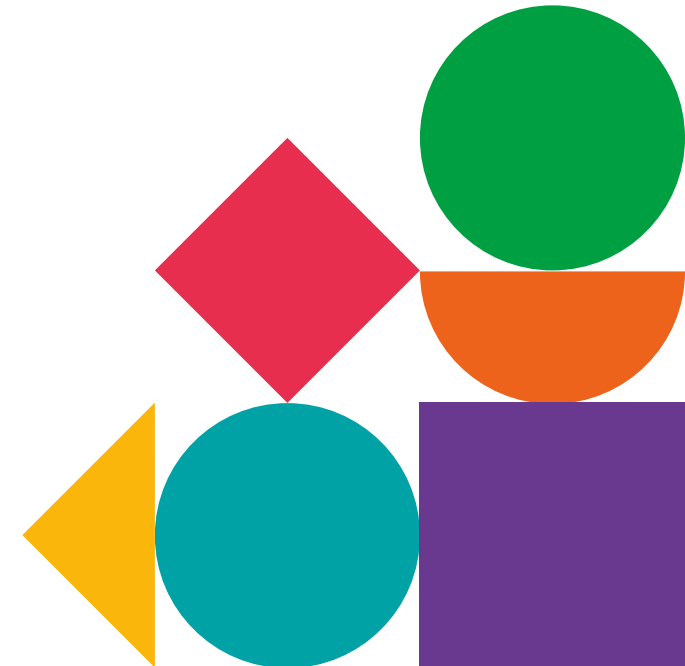
Challenge !!!
Solving conflicts



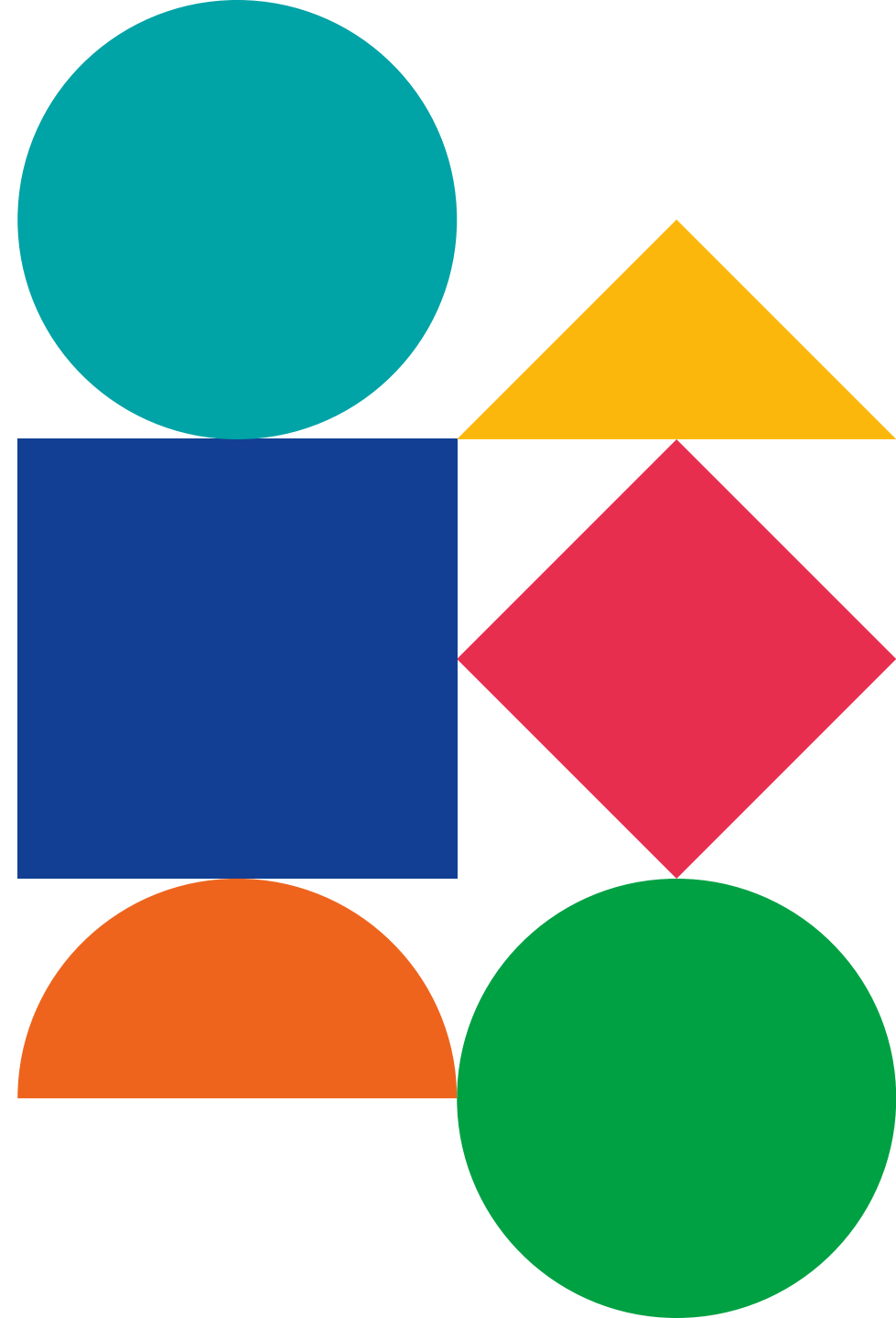
ACTIVITY

8. Merge one-by-one

- Merge all your branches with main
- Use conflict solving to keep only relevant information
- Think carefully of your commit messages
- Push your changes to GitHub



REVIEW



Review

```
user$ git branch <branch_name>
```

- Creates an alternative timeline

```
user$ git checkout <branch_name>
```

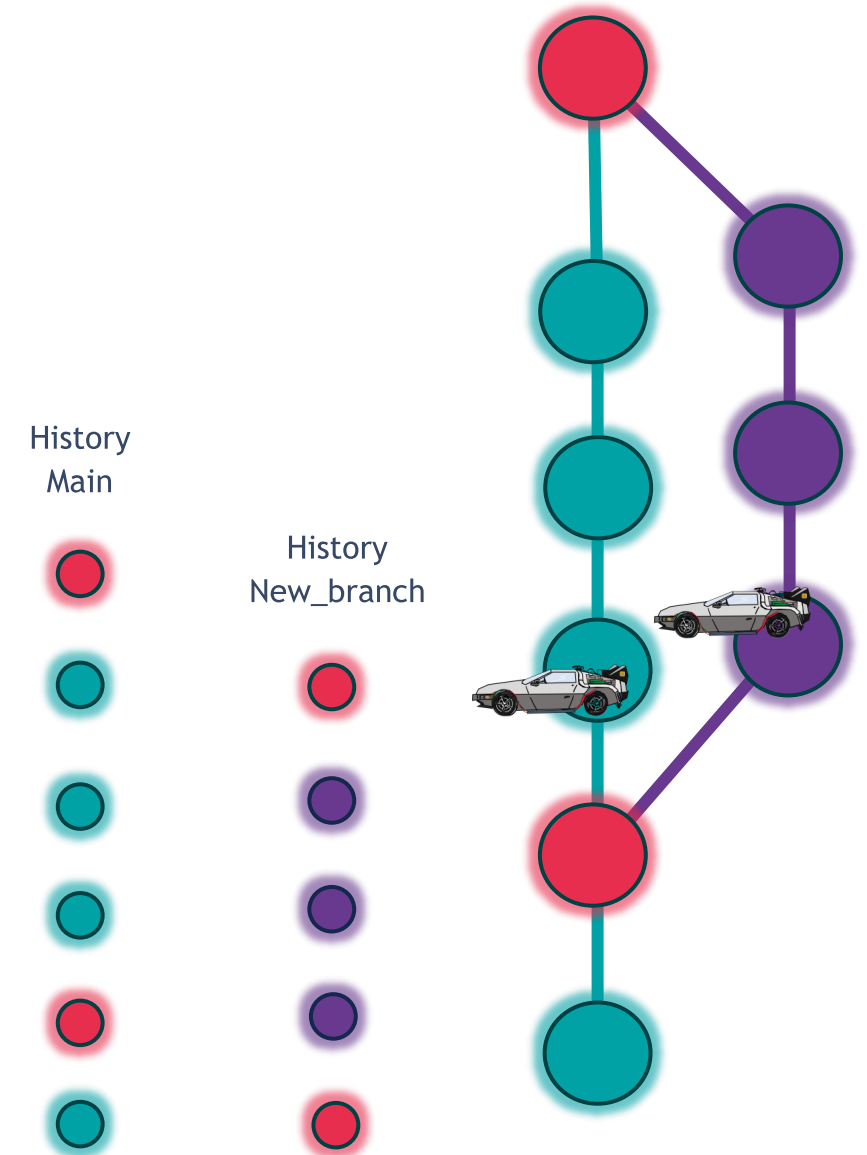
- Moves in the alternative timeline or old commit

```
user$ git checkout -b <branch_name>
```

- Creates and moves in the alternative timeline

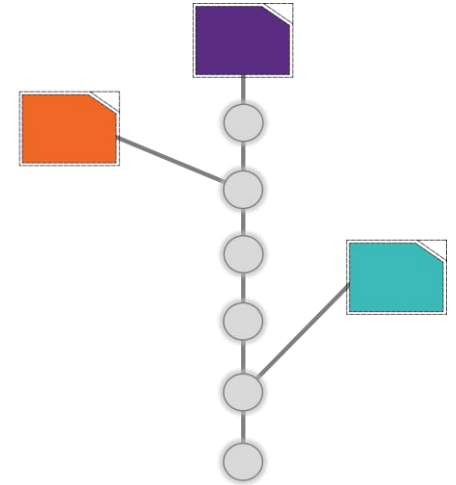
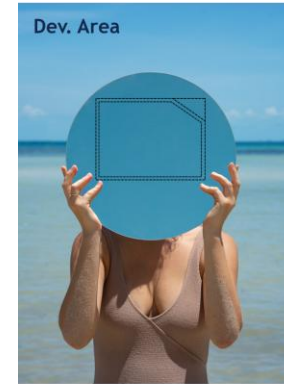
```
user$ git merge <branch_name>
```

- Merges two timelines in a shared commit point

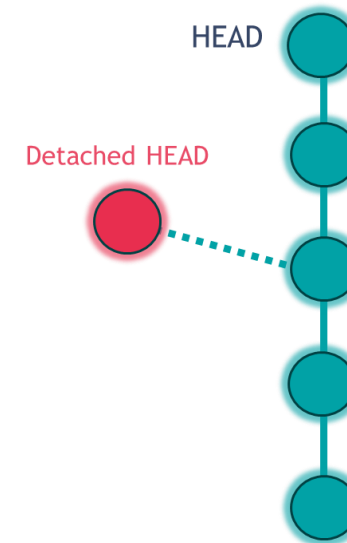


Review

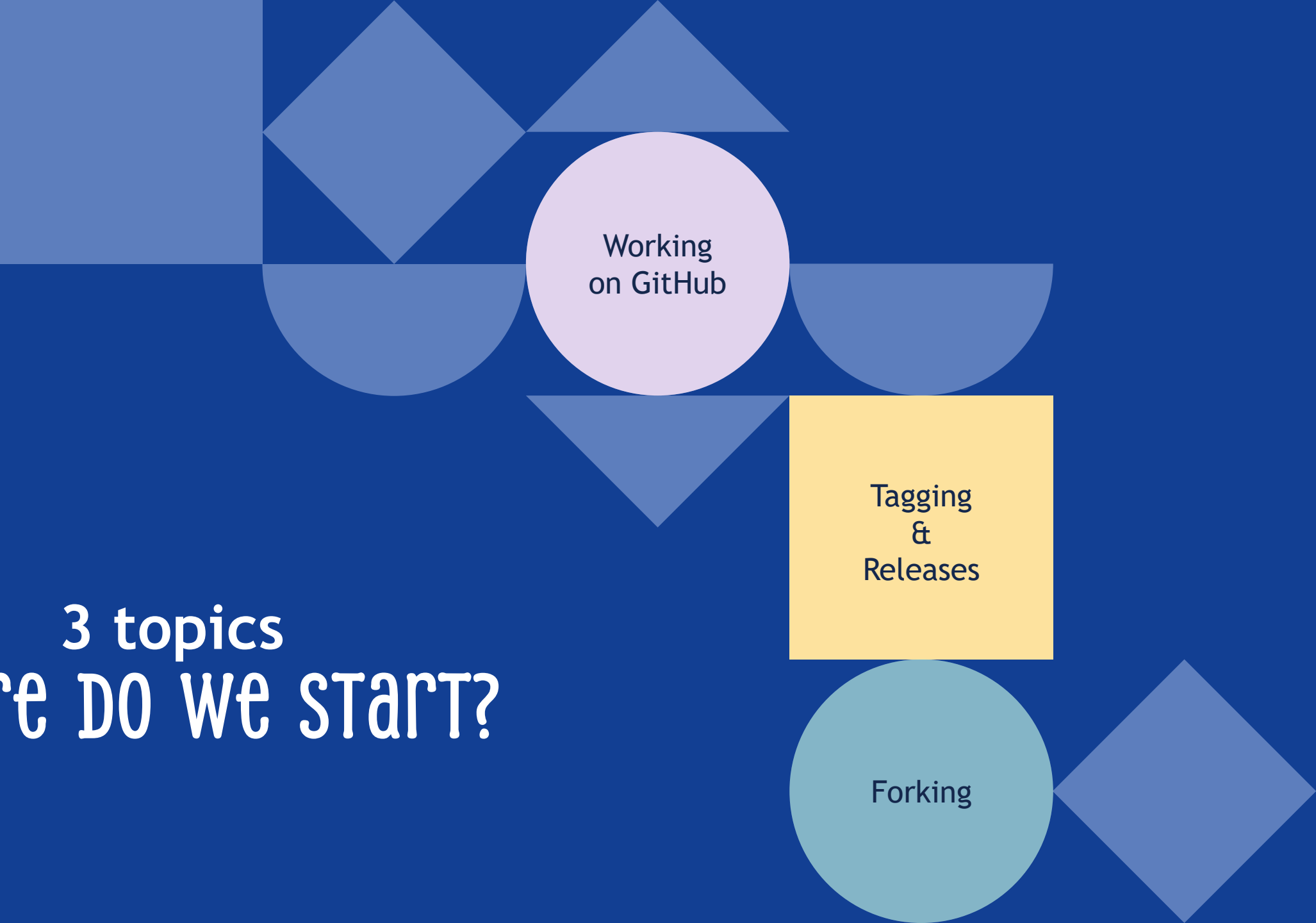
- The “mirror” effect
 - Folder Content == Branch you are working on



- Detached HEAD
 - Checking to old commit
 - A “floating” branch
 - Need to be connected



user\$ git switch -c <branch_name>

A diagram on a dark blue background with various geometric shapes in lighter shades of blue. A central light purple circle is connected by a downward-pointing triangle to a yellow square, which is then connected by another downward-pointing triangle to a light blue circle. The shapes are arranged in a vertical flow, with additional geometric shapes scattered around them.

Working
on GitHub

Tagging
&
Releases

Forking

3 topics
WHERE DO WE START?

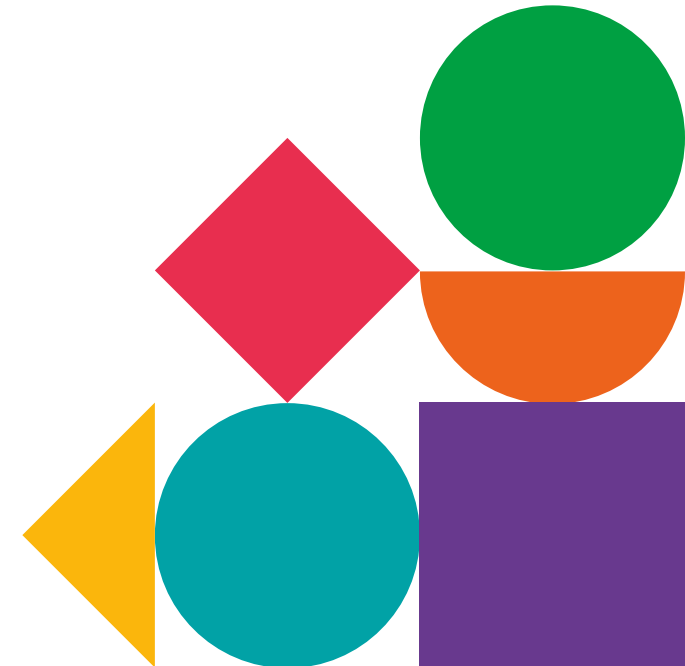


GitHub

Working
Online

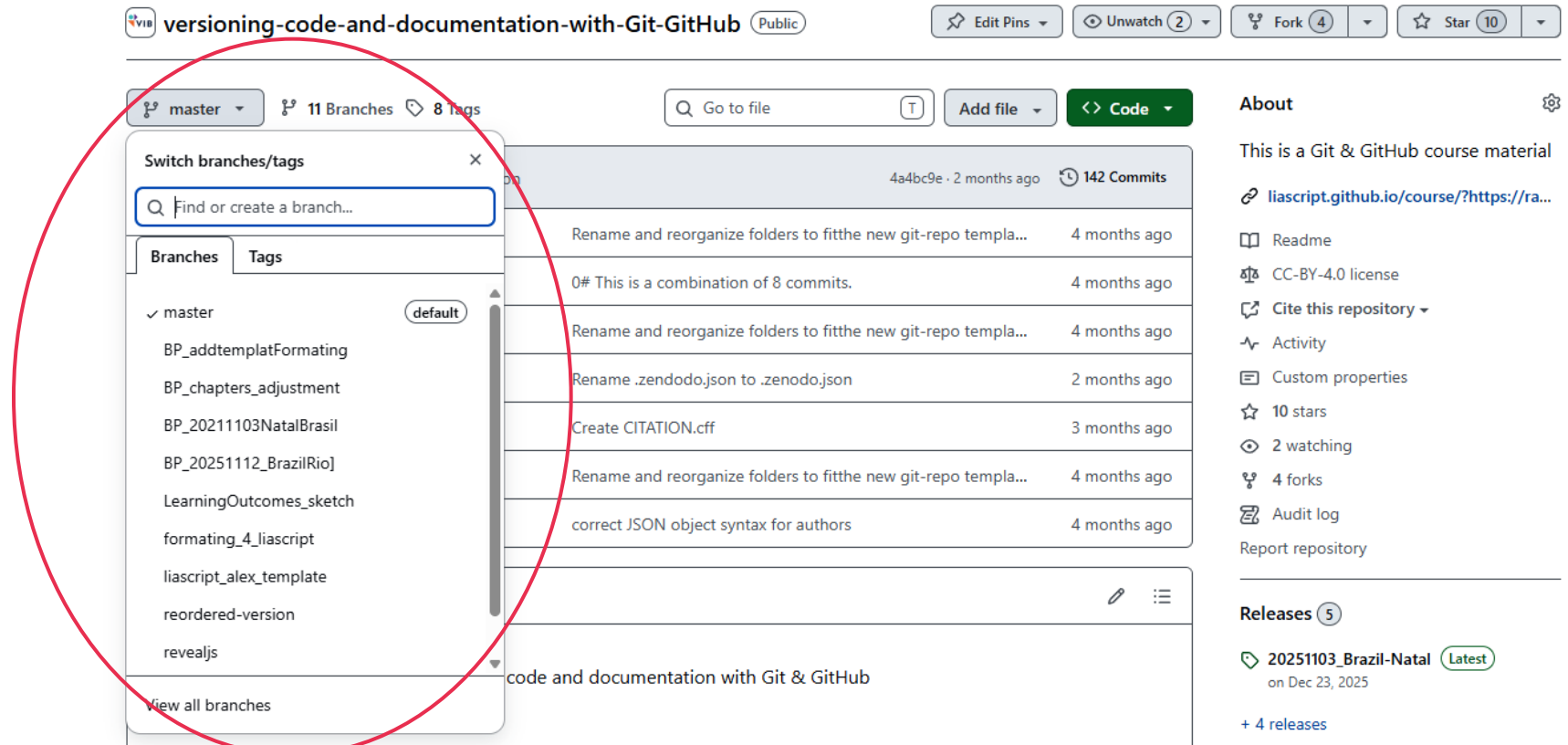
Basics on GitHub

- Check all branches
- Check all commits
- Editing
 - One-by-one
 - Several files
- Creating Issues
 - List of TODOs
 - List of bugs
- Tags and Releases



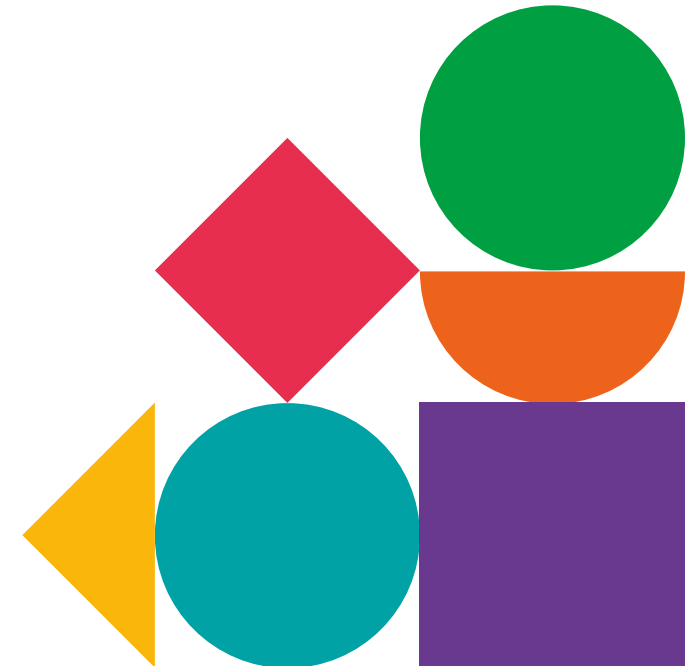
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Basics on GitHub

- Check all branches
- Check all commits

 abotzki	Rename .zendodo.json to .zenodo.json	4a4bc9e · 2 months ago	🕒 142 Commits
data	Rename and reorganize folders to fit the new git-repo template...	4 months ago	
docs	0# This is a combination of 8 commits.	4 months ago	
exercises	Rename and reorganize folders to fit the new git-repo template...	4 months ago	
.zenodo.json	Rename .zendodo.json to .zenodo.json	2 months ago	
CITATION.cff	Create CITATION.cff	3 months ago	
LICENSE	Rename and reorganize folders to fit the new git-repo template...	4 months ago	

1 file changed +24 -23 lines changed

Search within code

+24 -23

@@ -151,7 +151,7 @@ Authors

```
151 - @orcid(Bruna Piareck)](http://orcid.org/00000-0001-5958-0669)
152 - @orcid(James Collier)](http://orcid.org/0000-0002-0020-421X)
153 - @orcid(Tuur Muyldermans)](http://orcid.org/0000-0002-3926-7293)
154 - - @orcid(Alexander Botzki)](http://orcid.org/0000-0001-6691-4233)
```

Citing this lesson

@@ -171,121 +171,121 @@ There are preparation steps for all, Windows, Linux and Mac computers.

```
171
172 Most things you need to get installed before hands, one or the other are optional tools to help with
173 visualization and for somethings you can ask the trainer by mail or arriving 30 min before the class
174 starts.
```

```
174 - [click here ](https://liascript.github.io/course/?https://raw.githubusercontent.com/vib-tcp/introduction-github/refs/heads/0P\_addtemplatFormatting/docs/tutorials/01\_Get\_ready\_for\_the\_course.md#1)
```

```
151 - @orcid(Bruna Piareck)](http://orcid.org/00000-0001-5958-0669)
152 - @orcid(James Collier)](http://orcid.org/0000-0002-0020-421X)
153 - @orcid(Tuur Muyldermans)](http://orcid.org/0000-0002-3926-7293)
154 + - @orcid(Alexander Botzki)](http://orcid.org/0000-0001-6691-4233)
```

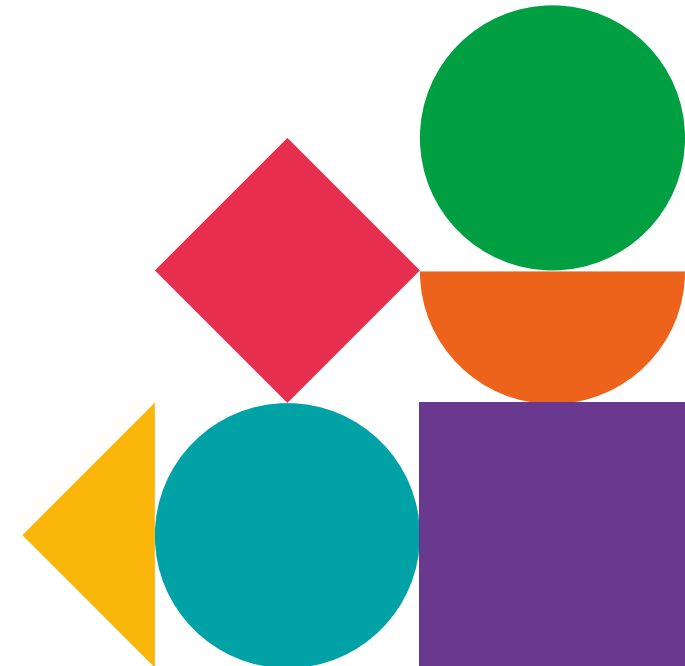
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```

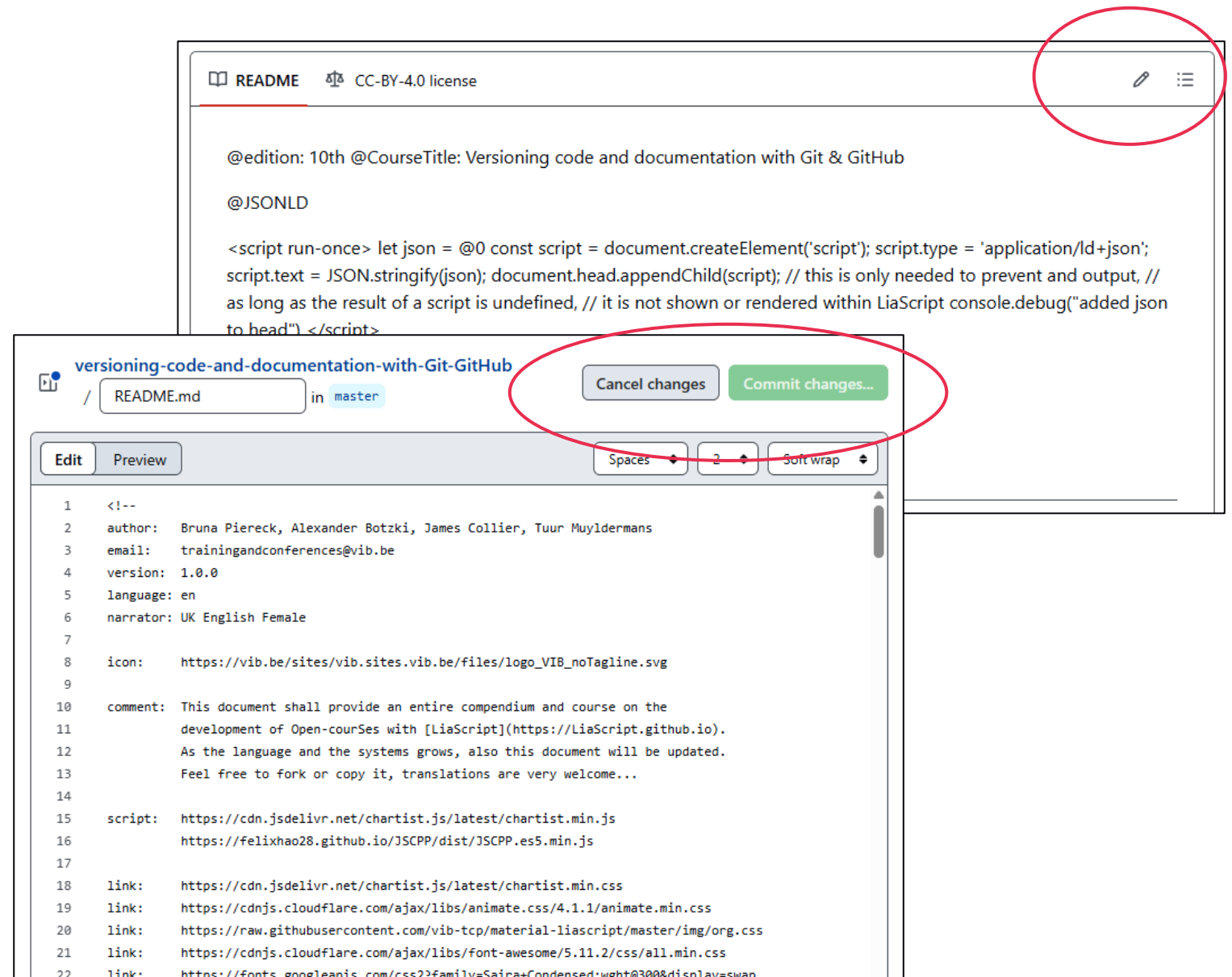
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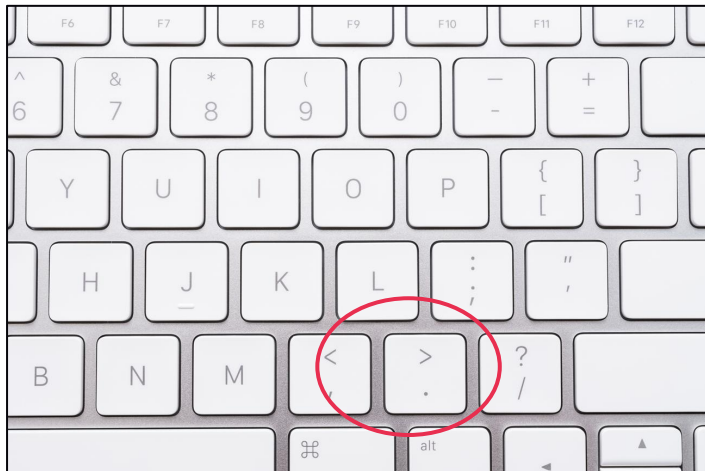
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Basics on GitHub

- Check all branches
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 - One-by-one
 - Several files



versioning-code-and-documentation-with-Git-GitHub Public

master 11 Branches 8 Tags

Go to file T Add file <> Code

Local Codespaces

Codespaces Your workspaces in the cloud + ...

No codespaces You don't have any codespaces with this repository checked out. Create codespace on master. Learn more about codespaces.

space usage for this repository is paid for by bpiereck.

space & GitHub

versioning-code-and-documentation-with-Git-GitHub / README.md

abotzki correct JSON object syntax for authors e0a0fa5 · 4 months ago

375 lines (247 loc) · 21.7 KB

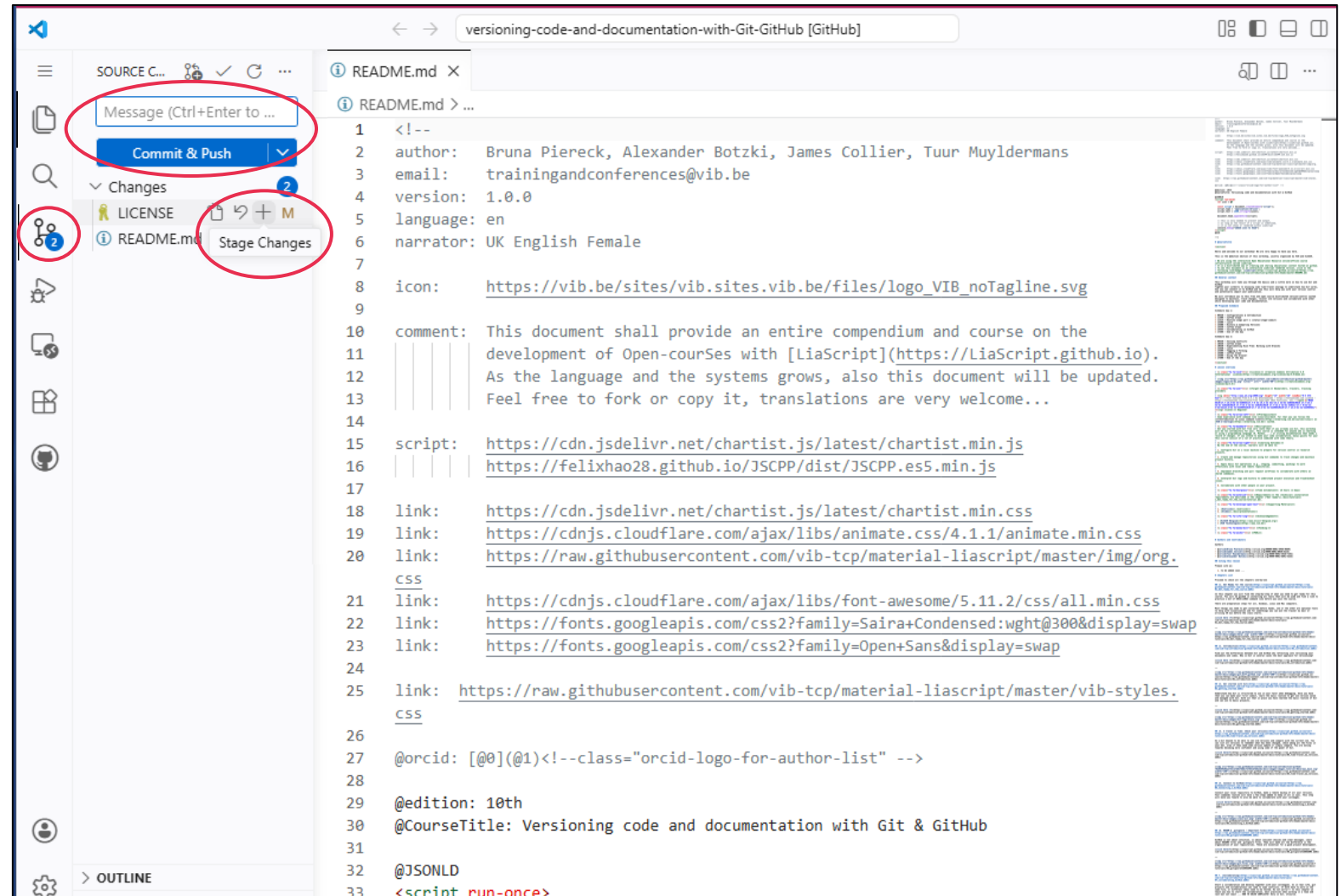
Preview Code | Blame

Edit file... In place (E) Open with... github.dev (selected) GitHub Desktop

```
@edition: 10th @CourseTitle: Versioning code and documentation with  
@JSONLD  
<script run-once> let json = @0 const script = document.createElement('application/ld+json'); script.text = JSON.stringify(json); document.head.appendChild(script);  
is only needed to prevent and output, // as long as the result of a script is  
shown or rendered within LiaScript console.debug("added json to head") </script>  
@end  
-->  
  
@CourseTitle
```

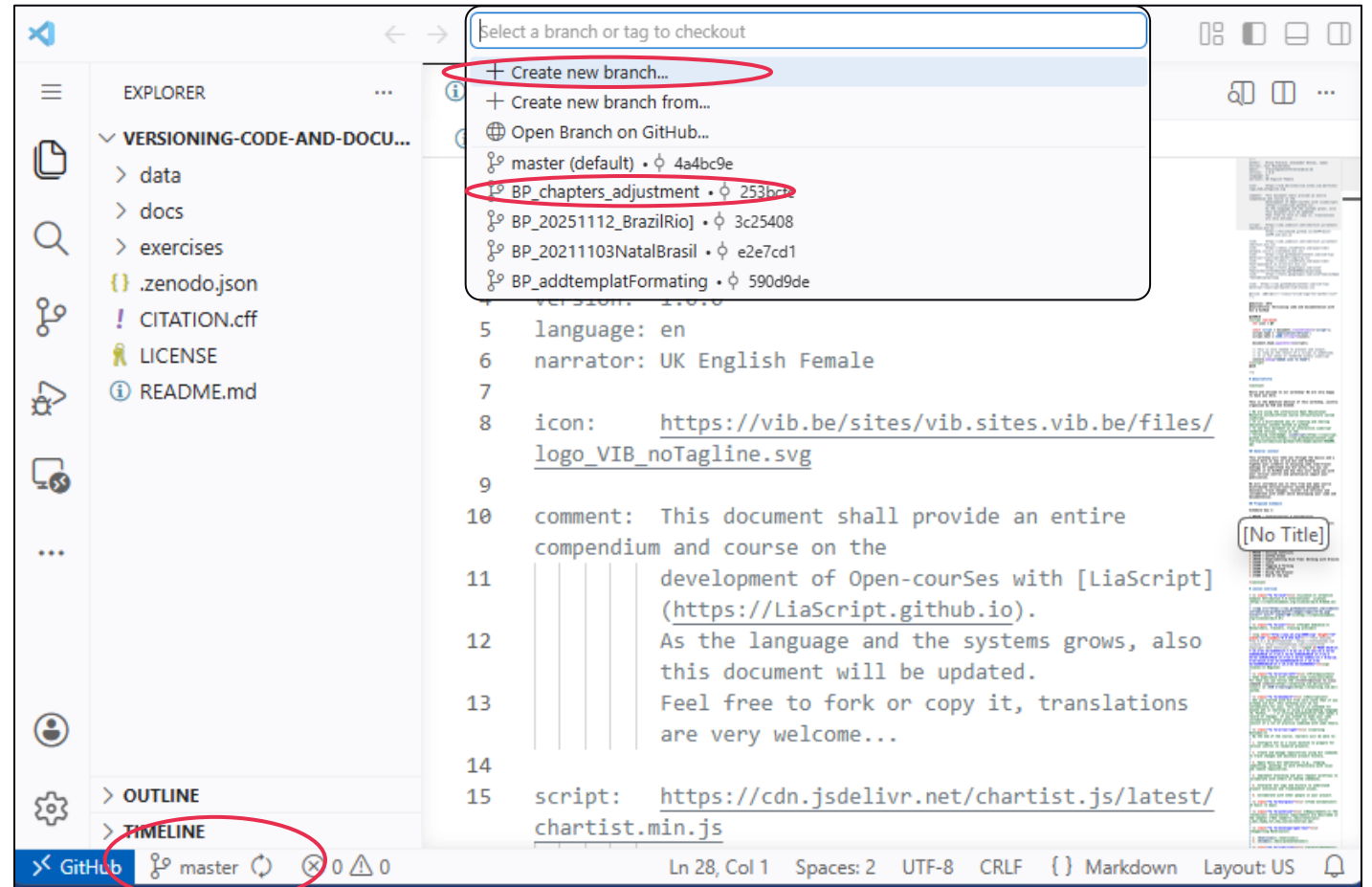
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Basics on GitHub

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Branches

Basics on GitHub

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The screenshot shows the GitHub interface for creating a new issue. The 'Issues' tab is highlighted with a red circle. The 'Create new issue' dialog is open, showing a form with a title field and a description field. The description field has a rich text editor toolbar. On the right side of the dialog, there are settings for Assignees, Labels, Type, Projects, and Milestone. At the bottom, there are buttons for 'Create more', 'Cancel', and 'Create'.

versioning-code-and-documentation-with-Git-GitHub

Code **Issues 7** Pull requests Agents Discussions Actions Projects Wiki Security and quality

Create new issue

Add a title *

Title

Add a description

Write Preview H B I | |

Type your description here...

Assignees

No one - [Assign yourself](#)

Labels

No labels

Type

No type

Projects

No projects

Milestone

No milestone

Paste, drop, or click to add files Write with Copilot

☐ Create more

Unwatch 2

<> Code

142 Commits

4 months ago

4 months ago

4 months ago

2 months ago

3 months ago

4 months ago

4 months ago

Basics on GitHub

- Check all branches
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 - List of bugs
- Tags and Releases

The screenshot shows a GitHub file view for a repository named "versioning-code-and-documentation-with-Git-GitHub". The file is "Git_QUIZ_intro2ndDay_En.md" and was last modified by user "bpiereck" 4 months ago. The interface includes tabs for "Preview", "Code", and "Blame", with "Code" being the active tab. A toolbar at the top right offers options like "Raw", "Copy", "Download", "Edit", and "Diff". The code content is a markdown file with a title, font specification, and a list of quiz questions. Line 10, which contains the text "b. Git is a website to keep backups of git projects", is highlighted in yellow. A red circle highlights the three-dot menu icon on the left of this line. Another red circle highlights the "Reference in new issue" option in the dropdown menu that appears. A third red circle highlights the "Copy permalink" option in the same menu. The bottom of the image shows the start of a new line with the command "c. `git commit -m 'project description'`".

versioning-code-and-documentation-with-Git-GitHub / exercises
/ Git_QUIZ_intro2ndDay_En.md

bpiereck Rename and reorganize folders to fit the new git-repo templ... bf42468 · 4 months ago History

Preview Code Blame Raw Copy Download Edit Diff

```
1 ---
2 title: "Git & GitHub intro QUIZ for 2nd day English version"
3 mainfont: "Liberation Mono"
4 ---
5
6 # 1. What is True about Git / GitHub?
7
8 a. GitHub is software for versioning
9
10 b. Git is a website to keep backups of git projects
11
12 backups and collaborations
13
14 c. Git is a website to keep backups of git projects
15
16 d. Git is a website to keep backups of git projects
17
18 e. Git is a website to keep backups of git projects
19
20 f. Git is a website to keep backups of git projects
21
22 g. Git is a website to keep backups of git projects
23
24 c. `git commit -m 'project description'`
```

Copy line
Copy permalink
View git blame
Reference in new issue
View file in GitHub.dev
View file in different branch/tag

Basics on GitHub

- Check all branches
- Check all commits
- Editing
 - One-by-one
 - Several files
- **Creating Issues**
 - **List of TODOs**
 - **List of bugs**
- Tags and Releases

Creating ISSUES:

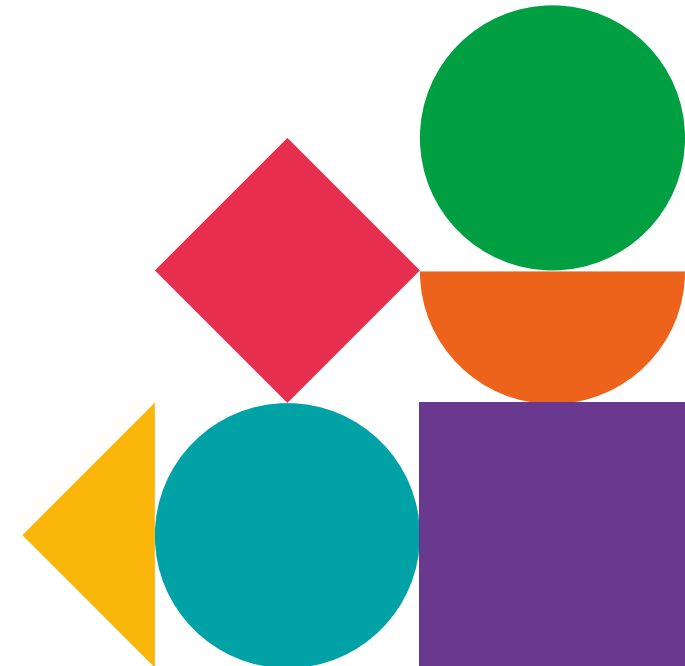
1. Descriptive title.
2. Explains the purpose of the issue
3. Include details that could help resolve the issue.

You can:

1. Create list of tasks
2. Create labels
3. Assign Issues to a collaborator
4. Exchange messages in a dedicated "chat"
5. Reference a line or paragraph

Basics on GitHub

- Check all branches
- Check all commits
- Editing
 - One-by-one
 - Several files
- Creating Issues
 - List of TODOs
 - List of bugs
- Tags and Releases



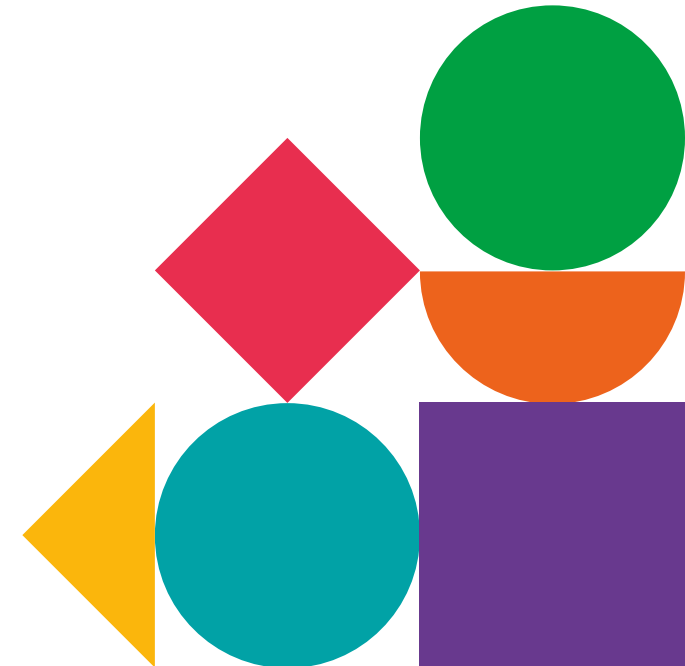
Tagging

Highlight
good versions

Tagging the commit

```
user$ git tag <name>
```

- Tag an important commit
 1. A good version
 2. A version for release
 3. A specific feature
 4. Experiment risk free
 5. Etc ..



Tagging the commit

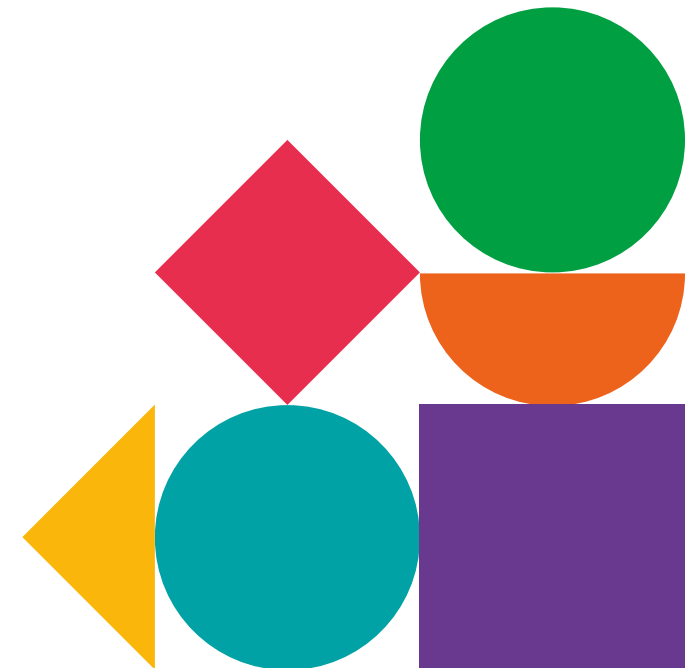
```
user$ git tag <name>
```

- Tag an important commit

1. A good version
2. A version for release
3. A specific feature
4. Experiment risk free
5. Etc ..

- Listing existing tags

```
user$ git tag  
user$ git tag -l  
user$ git tag --list
```



Tagging the commit

```
user$ git tag <name>
```

- Tag an important commit

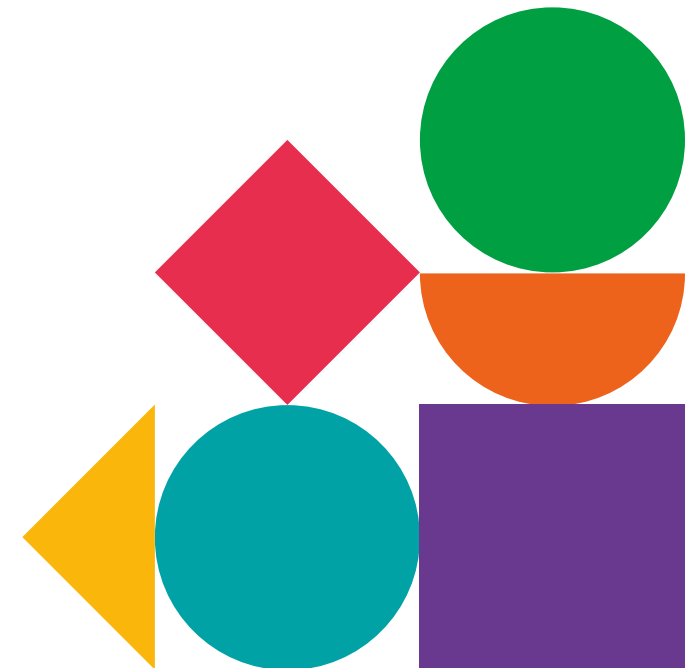
1. A good version
2. A version for release
3. A specific feature
4. Experiment risk free
5. Etc ..

- Listing existing tags

```
user$ git tag  
user$ git tag -l  
user$ git tag --list
```

- Tag an old commit

```
user$ git tag <name> <commit_ID>
```



ACTIVITY

9. Annotating

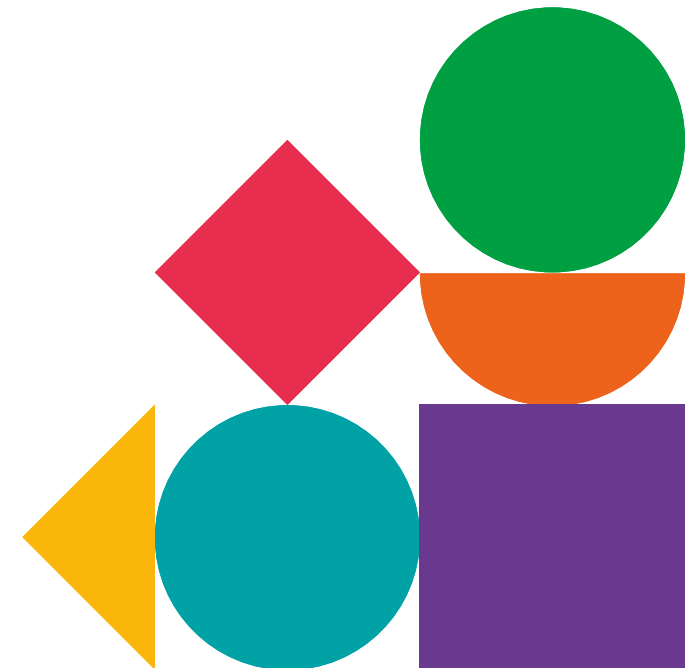
- Annotate in your documents about what is the tag used for
- What are the commands
- Commit

10. Creating tags

- Tag the last commit as “info_tag”
- Choose an old commit and create a tag for it
- List your tags

11. Challenge: Push your tags to GitHub

```
user$ git push --tags
```



Tags and GitHub releases

- Tag == commit “name”
- GitHub release

The screenshot shows the GitHub repository page for 'versioning-code-and-documentation-with-Git-GitHub'. The repository is public and has 11 branches and 8 tags. The '8 Tags' link is circled in red. Below the repository name, a table lists the files and their commit history:

File	Commit Message	Time Ago
data	Rename and reorganize folders to fithe new git-repo templa...	4 months ago
docs	0# This is a combination of 8 commits.	4 months ago
exercises	Rename and reorganize folders to fithe new git-repo templa...	4 months ago
.zenodo.json	Rename .zendodo.json to .zenodo.json	2 months ago
CITATION.cff	Create CITATION.cff	3 months ago
LICENSE	Rename and reorganize folders to fithe new git-repo templa...	4 months ago
README.md	correct JSON object syntax for authors	4 months ago

Below the file list, the README is visible, showing the course title and license. On the right side, the 'About' section provides details about the repository, including the license (CC-BY-4.0) and the number of stars (10). The 'Releases' section is circled in red, showing a single release named '20251103_Brazil-Natal' with the tag 'Latest'.

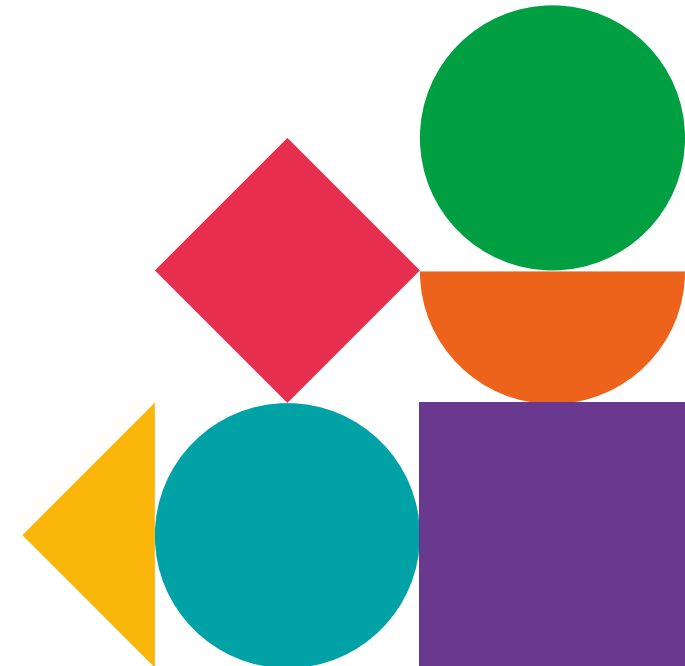
Tags and GitHub releases

- Tag == commit “name”
- GitHub release
 - Are based on tags
 - Allow for notes explanations
 - Can be sync to Zenodo for DOI

The screenshot displays a GitHub repository interface. At the top, the repository name 'versioning-code-and-documentation-with-Git-GitHub' is visible, along with navigation links like 'Edit Pins', 'Unwatch', 'Fork', and 'Star'. Below this, the repository's main navigation bar shows 'master' as the selected branch, '11 Branches', and '8 Tags' (the latter is circled in red). A search bar 'Go to file' and buttons for 'Add file' and 'Code' are also present. The repository description states 'This is a Git & GitHub course material'. The file explorer on the left lists files such as 'data', 'docs', 'exercises', '.zenodo.json', 'CITATION.cff', 'LICENSE', 'README.md', and 'README'. The 'Releases' tab is active, showing a list of releases. The first release, '20251112_BrazilRio', is highlighted, showing its date (Dec 23, 2025), author (bpierack), and description ('Course format that happen in Rio, funded by ELIXIR'). A 'Draft a new release' button is circled in red. A 'New release' modal is open, showing fields for 'Tag: Select tag', 'Target: master', 'Release title', and 'Release notes'. The 'Release notes' section includes a 'Generate release notes' button and a text area for describing the release. The bottom of the modal indicates 'Markdown is supported' and 'Paste, drop, or click to add files'.

ACTIVITY

12. Create a Release o GitHub

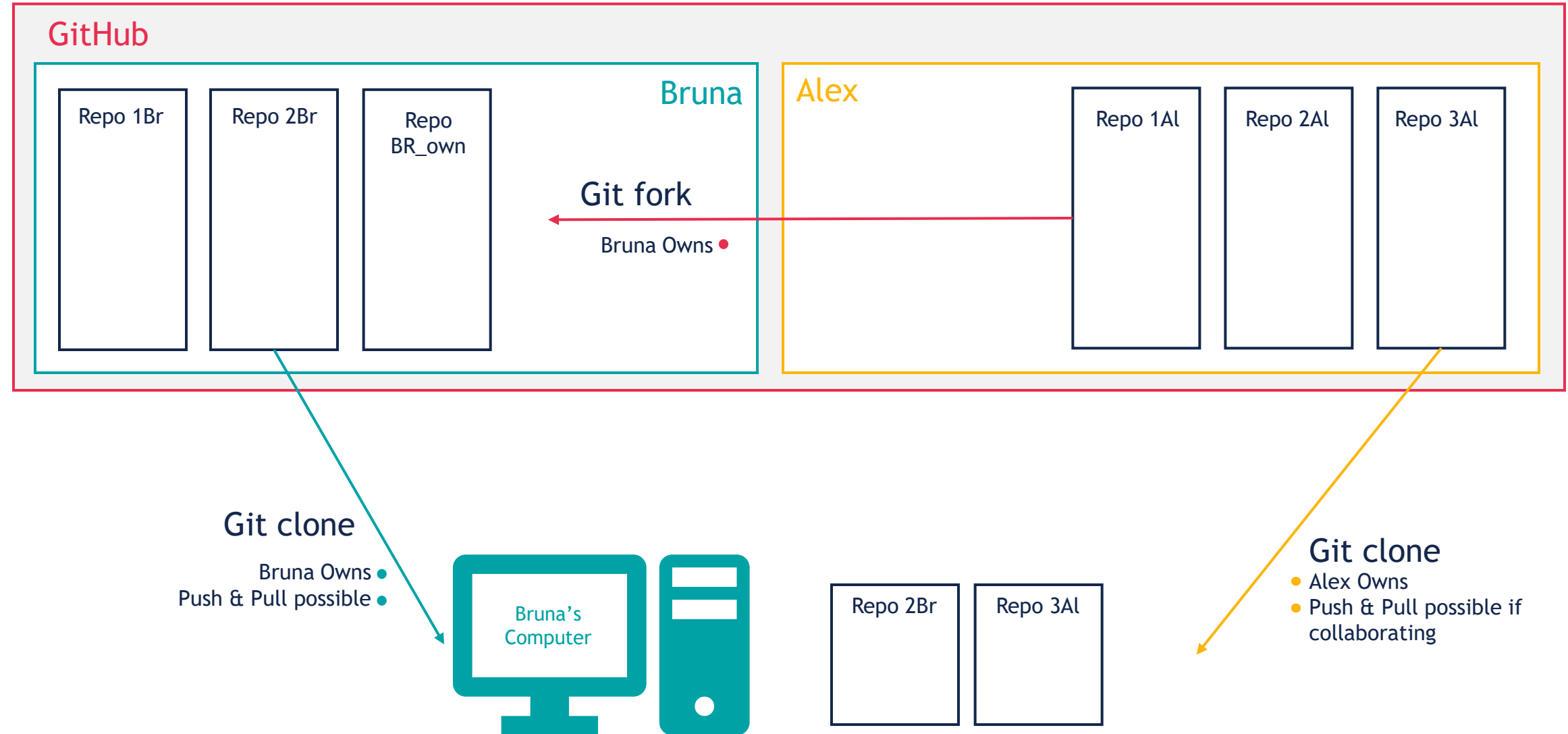


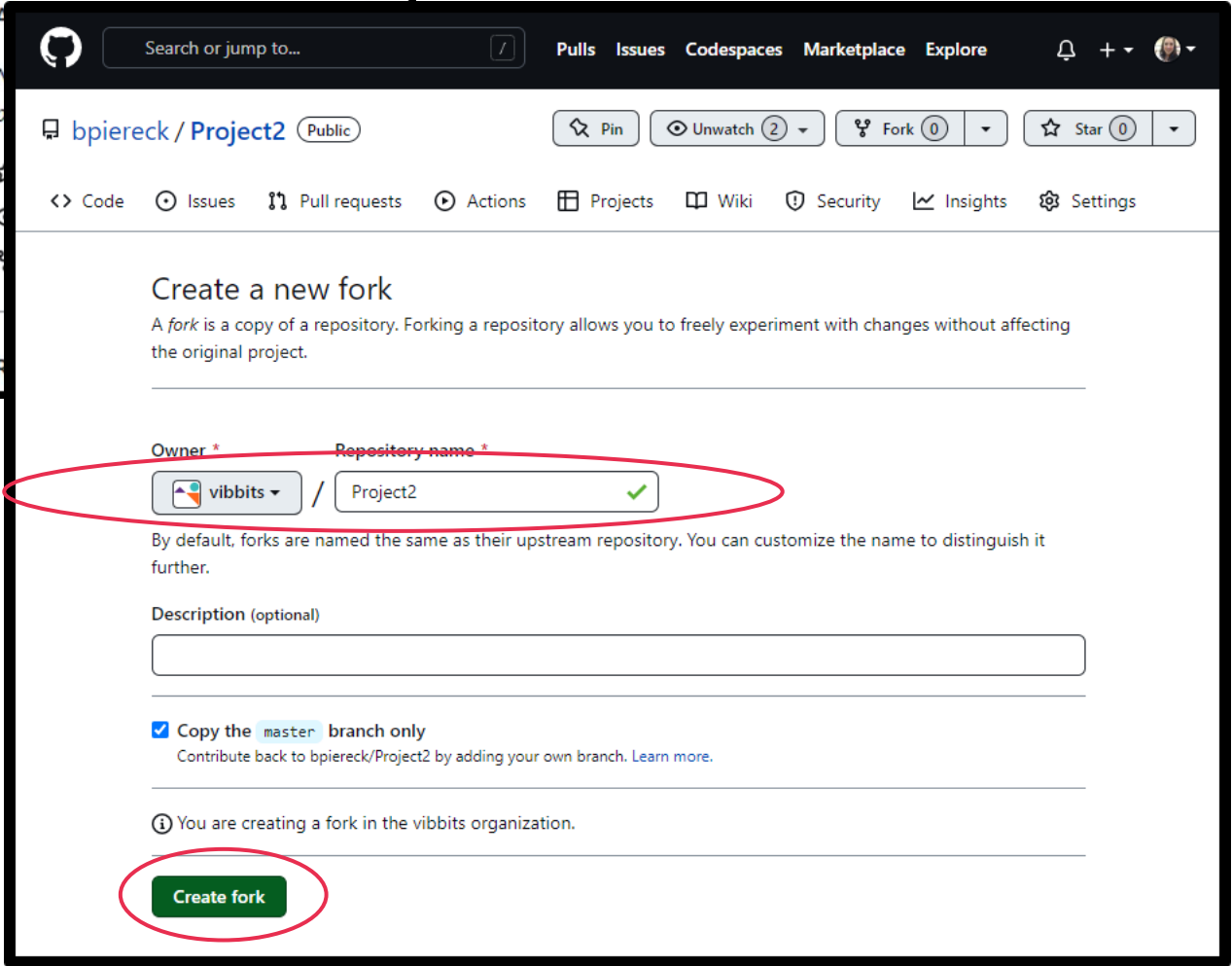
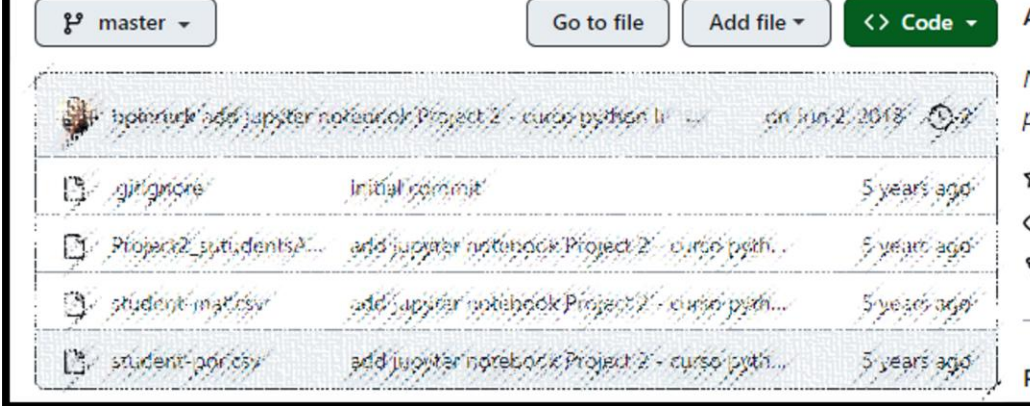
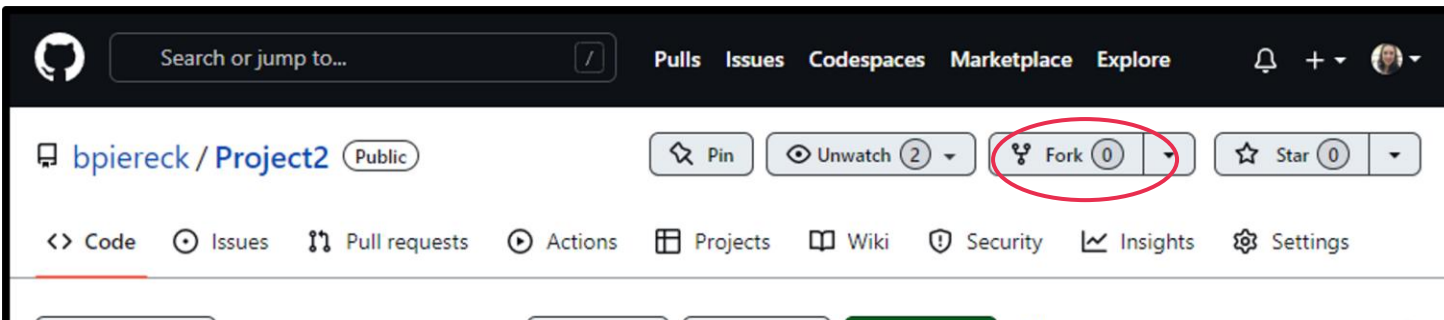
Forking



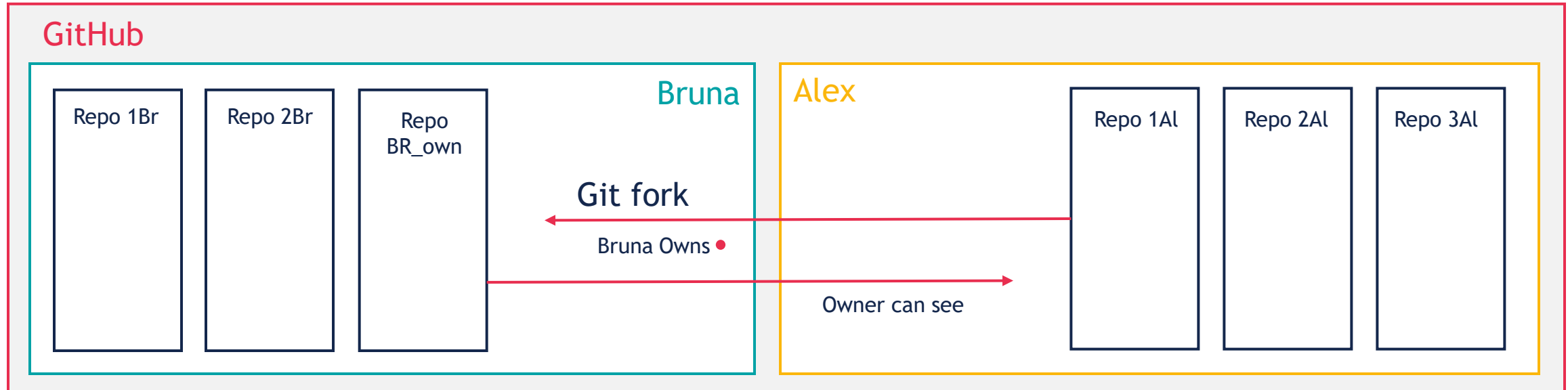
Copping
repositories

Fork vs Clone





Fork vs Clone



- Still connected
- Original creator can see it
- Changes can be incorporated
 - Both directions
 - Pull request

You are ready

In case of fire



1. git commit



2. git push

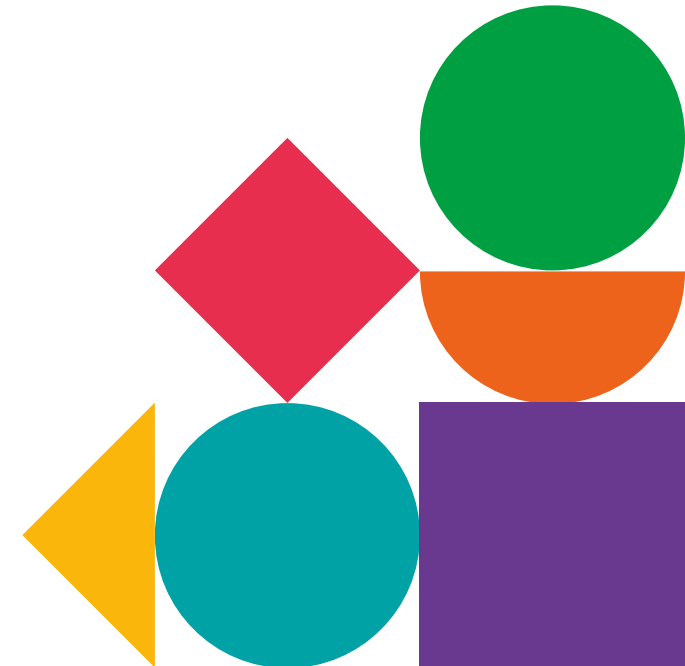


3. leave building

ACTIVITY

13. Fork race

- Wait for instructions to be a fair race!
- Add your name to the last line of the README file
- Commit (and wait)
- When instructed ask for a pull requested
 - A pull request must be evaluated and accepted





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